Capped GBM pricing

Replication, numerical convergence and parameter uncertainty

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11 October 2026

Abstract

I connect no-arbitrage replication, martingale valuation and numerical pricing for a capped European claim under geometric Brownian motion. A heat-coordinate transformation permits a monotone backward-Euler scheme with unconditional stability and proved convergence for the kinked payoff. Explicit domain and central-difference bounds separate pricing and Greek errors. Exact terminal simulation supplies pathwise, likelihood-ratio and bounded mixed-gamma estimators. Conditional expectation is derived as an L^2 projection, including jointly transformed controls and independent-pilot error. Synthetic Gaussian-return estimation distinguishes physical drift from risk-neutral pricing and derives exact parameter pivots. A capped price can have an interior volatility maximum: confidence propagation must evaluate this stationary point, while endpoint-only propagation has zero coverage at that maximum. Statistical and numerical confidence allowances are composed explicitly. Synthetic experiments retain adverse control rankings, failed coarse quadrature and coverage fluctuations. The analytic call spread remains the strongest known-parameter baseline. Detailed proofs and reproducible code support the mathematical and computational foundations of the basket-pricing project without establishing empirical model validity or practical quantum advantage.

Keywords: derivative pricing; geometric Brownian motion; replication; finite differences; numerical stability; convergence; Monte Carlo Greeks; conditional expectation; control variates; statistical calibration.
JEL classification: C15, C63, G13.

1 Financial model and no-arbitrage valuation

I use a one-asset claim to connect replication, martingale valuation, parabolic equations and statistical estimation under one financial model. The purpose is to demonstrate the mathematics underlying the numerical comparisons. The classical valuation framework follows Black and Scholes (1973) and Merton (1973); these established methods do not establish a quantum advantage or a new financial theory. Replication and call valuation specialise Black and Scholes (1973); the continuous-dividend treatment follows Merton (1973). The pricing and sensitivity derivations below retain the assumptions required for these specialisations.

Let $S > 0$, $K > 0$, $H > 0$, $\sigma > 0$ and $\tau > 0$. The interest rate r and continuous dividend yield q are constant. Write $J = K + H$ and

$$F(x) = \min\{(x-K)^+, H\} = (x-K)^+ - (x-J)^+, \qquad D = \exp(-r\tau), \qquad v = \sigma\sqrt{\tau}. \tag{1}$$

Under the selected pricing measure Q, the ex-dividend asset follows

$$dS_t = (r-q)S_t\,dt + \sigma S_t\,dW_t^Q, \qquad X = S\exp\{b\tau + vZ\}, \qquad b = r - q - \tfrac{1}{2}\sigma^2, \tag{2}$$

where Z is standard normal. The price is

$$V(S,\tau;r,q,\sigma) = D\,\mathbb{E}_Q[F(X)]. \tag{3}$$

This is a model comparison with synthetic parameters, rather than a calibrated empirical test. A physical return mean cannot be substituted for the pricing drift $r - q$. I assume frictionless continuous trading, reinvestable dividends, borrowing and lending at the same rate r, and an unrestricted stock position within the stated admissible strategy class. The probability space carries the usual augmented Brownian filtration, satisfying the usual completeness and right-continuity conditions.

Proposition 1.1 (Replication and the pricing equation)**.** *The price in* (3) *is bounded, continuous at expiry and smooth for $S, \tau > 0$. It satisfies*

$$V_\tau = \tfrac{1}{2}\sigma^2 S^2 V_{SS} + (r-q)SV_S - rV, \qquad V(S,0) = F(S). \tag{4}$$

Holding V_S shares and placing $V - SV_S$ in the money-market account replicates the claim when dividends are reinvested within the self-financing strategy. Its discounted wealth is non-negative, and its stochastic gain is square-integrable on each finite horizon. Under the usual admissibility condition that discounted wealth is bounded below, the stipulated market has no arbitrage. This replicating price is the unique no-arbitrage price among bounded claim valuations and the unique bounded classical solution of the terminal-value problem. Here a classical solution is $C^{1,2}$ for positive remaining maturity and is continuous at expiry.

The dividend term belongs in the self-financing gain equation, not in the terminal payoff. Appendix A derives both the measure change and the replication argument, including the limiting argument at the payoff kinks.

2 Analytic prices and Greeks

Let Φ and φ denote the standard normal distribution function and density. For strike $A > 0$, define

$$d_{1,A} = \frac{\log(S/A) + (r - q + \sigma^2/2)\tau}{v}, \qquad d_{2,A} = d_{1,A} - v, \qquad C_A = Se^{-q\tau}\Phi(d_{1,A}) - AD\Phi(d_{2,A}). \tag{5}$$

Then $V = C_K - C_J$. Write $\Delta\Phi_i = \Phi(d_{i,K}) - \Phi(d_{i,J})$ and $\Delta\varphi = \varphi(d_{1,K}) - \varphi(d_{1,J})$. Differentiation gives

$$\Delta = V_S = e^{-q\tau}\Delta\Phi_1, \tag{6}$$

$$\Gamma = V_{SS} = \frac{e^{-q\tau}\Delta\varphi}{Sv}, \tag{7}$$

$$\mathcal{V} = V_\sigma = Se^{-q\tau}\sqrt{\tau}\,\Delta\varphi, \tag{8}$$

$$\mathcal{R} = V_r = \tau D\{K\Phi(d_{2,K}) - J\Phi(d_{2,J})\}, \tag{9}$$

$$\Theta = -V_\tau = -\frac{Se^{-q\tau}\sigma}{2\sqrt{\tau}}\Delta\varphi + qSe^{-q\tau}\Delta\Phi_1 - rD\{K\Phi(d_{2,K}) - J\Phi(d_{2,J})\}. \tag{10}$$

Vega and rho are derivatives per unit change in volatility and interest rate. Calendar theta uses a fixed expiry and advances the valuation date, so it is the negative maturity derivative. Gamma and vega can be negative for a capped claim; this is a property of the call spread, not a numerical failure.

3 A monotone heat-equation discretisation

Introduce $y = \log(S/K) + b\tau$, $a = \sigma^2/2$ and $f(y) = F(Ke^y)$. The whole-line heat solution w satisfies

$$w_\tau = aw_{yy}, \qquad w(0,y) = f(y), \qquad V(S,\tau) = Dw(\tau, y). \tag{11}$$

Its payoff kinks lie at 0 and $c = \log(J/K)$. Moreover, $0 \le f \le H$ and f is globally Lipschitz with constant $L_f = J$. For a finite domain $L < 0 < c < U$, let w^{box} solve the same equation with fixed boundary values 0 and H. These values approximate the exact financial far-field limits

$$\lim_{S\downarrow 0} V(S,\tau) = 0, \qquad \lim_{S\to\infty} V(S,\tau) = DH. \tag{12}$$

Indeed the terminal asset tends pointwise to zero or infinity as its initial spot does, and the bounded continuous payoff permits dominated convergence. At remaining maturity τ, the heat-domain endpoints correspond to asset values $K\exp(L - b\tau)$ and $K\exp(U - b\tau)$. Their finite values do not make the imposed boundary prices exact; the exit-tail bound below quantifies the approximation.

For integers $M \ge 3$ and $N \ge 1$, set $h = (U - L)/M$, $y_j = L + jh$, $k = \tau/N$ and $\lambda = ak/h^2$. Backward Euler gives

$$-\lambda W_{j-1}^n + (1 + 2\lambda)W_j^n - \lambda W_{j+1}^n = W_j^{n-1}, \qquad W_0^n = 0, \quad W_M^n = H, \quad W_j^0 = f(y_j). \tag{13}$$

I interpolate the final grid values linearly at the query y and then apply the discount factor. Subtracting the affine boundary lift diagonalises this same finite-difference operator in the discrete sine basis. Its eigenvalues and backward-Euler factors are

$$\nu_\ell = \frac{4}{h^2}\sin^2\left(\frac{\ell\pi}{2M}\right), \qquad (1 + ak\nu_\ell)^{-N}, \qquad 1 \le \ell \le M - 1. \tag{14}$$

This is a faster evaluation of (13), rather than a different spatial method. A direct tridiagonal time-step implementation checks that equivalence numerically. A floating-point implementation is not an interval-arithmetic certificate merely because its mathematical scheme has a proved bound.

Theorem 3.1 (Unconditional stability and a nonsmooth-payoff error bound)**.** *For every $h, k > 0$,* (13) *preserves $0 \le W_j^n \le H$. Differences between two solutions with the same boundary values contract in the maximum norm at every step. Fix an even, non-negative $\rho \in C_c^\infty(-1,1)$ with integral one, and set*

$$m_1 = \int_{-1}^{1} |z|\rho(z)\,dz, \qquad c_3 = \|\rho'''\|_{L^1}, \qquad 0 < \eta < \min\{-L, U - c\}. \tag{15}$$

For exact arithmetic, the nodal error at maturity obeys

$$\max_{1\le j\le M-1} |W_j^N - w^{\mathrm{box}}(\tau, y_j)| \le 2L_f m_1 \eta + \tau\left(\frac{ah^2}{12} + \frac{a^2k}{2}\right) L_f c_3 \eta^{-3}. \tag{16}$$

Consequently the scheme converges to the finite-domain solution, with a rigorous upper bound of order $(h^2 + k)^{1/4}$ for this nonsmooth payoff on a fixed time interval. This conservative bound does not assert that the observed rate is exactly one quarter.

The proof uses the maximum principle, a compact mollifier and an odd periodic extension after the boundary lift. It does not assume that the unmollified payoff has four derivatives. The full proof is in Appendix B.

Proposition 3.2 (Domain and interpolation errors)**.** *For $x \in [L, U]$, put*

$$\beta(\tau, x) = \min\left\{1, 2\Phi\left(-\frac{x - L}{\sigma\sqrt{\tau}}\right) + 2\Phi\left(-\frac{U - x}{\sigma\sqrt{\tau}}\right)\right\}. \tag{17}$$

Then $|w(\tau,x) - w^{\mathrm{box}}(\tau,x)| \le H\beta(\tau,x)$. *Let* y *lie between grid nodes* y_j, y_{j+1} *and define* $\beta_* = \max\{\beta(\tau,y_j), \beta(\tau,y_{j+1})\}$. *The interpolated, discounted price* $\widehat{V}$ *satisfies*

$$|\widehat{V} - V| \le D\left[H\beta_* + 2L_f m_1 \eta + \tau\left(\frac{ah^2}{12} + \frac{a^2 k}{2}\right) L_f c_3 \eta^{-3} + \frac{h^2 L_f}{8\sigma\sqrt{\tau}}\sqrt{\frac{2}{\pi}}\right]. \tag{18}$$

The four terms control the finite domain, payoff smoothing, numerical consistency and interpolation. Payoff smoothing is a proof device; the implemented initial condition remains the original capped payoff. On a fixed domain, mesh refinement converges to w^{box}, leaving a domain error relative to the whole-line financial price. Full-price consistency requires the domain error to vanish as well.

4 Greek consistency and Monte Carlo comparisons

Nonsmooth payoffs require separate control of sensitivities; the analysis of Giles and Carter (2006) gives a published comparison for Crank–Nicolson/ Rannacher schemes. I prove the bounds for the backward-Euler scheme used here.

I compute PDE Greeks by central price differences on the same domain and grid. This fixes the numerical heat domain under a bump; the transformed query y, diffusivity a and discount factor still change with the bumped parameter. The small-bump limit must respect the price error.

Proposition 4.1 (Propagation into central differences)**.** *Suppose* $|\widehat{V} - V| \le E$ *at every required bumped point. For a scalar parameter* p, *if* $|\partial_p^3 V| \le M_3$ *on its bump interval, then*

$$\left|\frac{\widehat{V}(p+d) - \widehat{V}(p-d)}{2d} - \partial_p V(p)\right| \le \frac{E}{d} + \frac{M_3 d^2}{6}. \tag{19}$$

For spot gamma, if $|\partial_S^4 V| \le M_4$ *on the spot bump interval, then*

$$\left|\frac{\widehat{V}(S+d) - 2\widehat{V}(S) + \widehat{V}(S-d)}{d^2} - \Gamma\right| \le \frac{4E}{d^2} + \frac{M_4 d^2}{12}. \tag{20}$$

These statements apply uniformly on a compact parameter set with S, σ, τ *bounded away from zero, provided all queries lie inside the domain. Refining the mesh and enlarging the domain so that* $E_n \to 0$, *then choosing* $d_n \to 0$ *and* $E_n/d_n \to 0$ *for first derivatives or* $E_n/d_n^2 \to 0$ *for gamma, proves consistency for the whole-line Greeks.*

After fixing units, choices proportional to $E_n^{1/3}$ and $E_n^{1/4}$ balance the displayed error terms for first derivatives and gamma, respectively. Practical comparisons also retain the exact analytic central-difference bias at each implemented bump. This diagnoses numerical price error separately from bump bias and prevents an apparent Greek discrepancy from being assigned wholly to the PDE scheme.

The implemented bump recipe uses a dimensionless log-grid spacing h, a positive spot reference scale S_*, and positive reference scales for the other parameters. The spot bump is proportional to $S_* h^{1/8}$; first-order parameter bumps are proportional to $h^{1/4}$ in their respective reference units. With $k = O(h^2)$, the rigorous price bound is $O(h^{1/2})$ after making the domain error $o(h^{1/2})$. The resulting spot gamma error is $O(h^{1/4})$, and the first-order errors are at most $O(h^{1/4})$. These are conservative guaranteed upper rates, not fitted empirical orders. A finite fixed domain retains its displayed exit-tail allowance, regardless of this bump recipe.

Pathwise and likelihood-ratio estimators have established foundations (Broadie and Glasserman, 1996). The capped-payoff formulae and their ranges are derived explicitly in Appendix C.

For exact terminal sampling, write $I = \mathbf{1}_{\{K<X<J\}}$ and define the following one-path estimators:

$$Y_V = DF(X), \qquad Y_\Delta = DIX/S,$$
$$Y_\mathcal{V} = DIX(\sqrt{\tau}Z - \sigma\tau), \qquad Y_\mathcal{R} = D\tau\{IX - F(X)\}, \tag{21}$$
$$Y_\Theta = D\left[rF(X) - IX\left(b + \frac{\sigma Z}{2\sqrt{\tau}}\right)\right], \tag{22}$$
$$Y_\Gamma = \frac{DF(X)}{S^2}\left\{\frac{Z^2-1}{v^2} - \frac{Z}{v}\right\}. \tag{23}$$

The first-order Greeks use pathwise differentiation, following the simulation sensitivity framework of Broadie and Glasserman (1996). Gamma uses the second likelihood-ratio weight: differentiating the pathwise indicator as though it were smooth would omit its boundary contribution. I also retain the bounded mixed pathwise/likelihood-ratio rival

$$Y_{\Gamma,\mathrm{mix}} = \frac{DIX}{S^2}\left(\frac{Z}{v} - 1\right). \tag{24}$$

It follows by differentiating delta's fixed-asset density representation, rather than differentiating a random indicator pointwise. Its derivation appears in Appendix C. Boundedness gives a different finite-sample uncertainty guarantee, but does not by itself prove variance dominance over the ordinary likelihood-ratio estimator.

Proposition 4.2 (Unbiased estimators and finite-sample uncertainty)**.** *Each estimator in* (21)–(24) *has expectation equal to its analytic price or Greek and finite second moment. Let*

$$z_- = \frac{\log(K/S) - b\tau}{v}, \quad z_+ = \frac{\log(J/S) - b\tau}{v}, \quad m = \max\{|z_-|, |z_+|\}. \tag{25}$$

Deterministic ranges for price, delta, vega, rho and theta are, respectively,

$$[0, DH], \quad [0, DJ/S], \quad [-B_\mathcal{V}, B_\mathcal{V}], \quad [-D\tau H, D\tau K], \quad [-B_\Theta, B_\Theta],$$
$$B_\mathcal{V} = DJ(\sqrt{\tau}m + \sigma\tau), \qquad B_\Theta = D\left[|r|H + J\left(|b| + \frac{\sigma m}{2\sqrt{\tau}}\right)\right]. \tag{26}$$

For n independent raw paths and a range width R, a distribution-free two-sided confidence radius with failure probability at most α, following Hoeffding (1963), is

$$R\sqrt{\frac{\log(2/\alpha)}{2n}}. \tag{27}$$

For gamma, the bound

$$\mathrm{Var}(Y_\Gamma) \le B_\Gamma^2 := \frac{D^2H^2}{S^4}\left(\frac{2}{v^4} + \frac{1}{v^2}\right) \tag{28}$$

gives the finite-sample Chebyshev radius $B_\Gamma/\sqrt{n\alpha}$. The mixed gamma estimator has range

$$[-B_{\mathrm{mix}}, B_{\mathrm{mix}}], \qquad B_{\mathrm{mix}} = \frac{DJ}{S^2}\left(\frac{m}{v} + 1\right), \tag{29}$$

so it admits the Hoeffding radius with width $2B_{\mathrm{mix}}$. Allocating failure probabilities across outputs and using a union bound gives a simultaneous statement. Sample-variance normal intervals are separate asymptotic diagnostics and do not have these finite-sample coverage guarantees.

Contract	S	H	τ	σ	r	q
Lower kink	100	20	1	0.2	0.03	0.01
Upper kink	120	20	1	0.2	0.03	0.01
Inside spread	110	20	1	0.2	0.03	0.01
Deep cap	160	20	1	0.2	0.03	0.01
Low-volatility short dated	100	20	0.05	0.06	0.03	0.01
Negative rate/dividend	105	20	1	0.25	-0.015	0.045
High volatility	95	30	2	0.65	0.03	0.01

Table 1: Wholly synthetic inputs. Rates, volatility and dividend yields are expressed per year; the cap and spot use common monetary units. The labels describe designed test cases rather than an empirical asset sample.

The numerical study allocates 0.05/49 to each of the six ordinary raw outputs and the additional raw mixed gamma across seven contracts. The corresponding 49-interval family has failure probability at most 0.05 under the ideal independent-normal sampling model. Comparisons between Monte Carlo methods additionally report sample-variance normal intervals as asymptotic diagnostics.

The strongest implemented Monte Carlo comparison combines antithetic normal pairs with a centred discounted-stock control, $DX - Se^{-q\tau}$. I fit each control coefficient on independent pilot pairs and hold it fixed for the main sample. Conditional on the pilot, the controlled mean is unbiased because the control has expectation zero. Each independent pair, rather than its two correlated paths, is the sampling unit. The pilot and both terminal evaluations per pair belong in the computational work. The controlled estimators contain an unbounded lognormal control, so their sample-variance intervals remain asymptotic; the bounded raw-path Hoeffding guarantees are not transferred to them.

5 Synthetic comparison design and numerical evidence

I use seven designed contracts, all with $K = 100$, to exercise both payoff kinks, the interior spread, deep capping, a short low-volatility horizon, negative interest with dividends and a volatile long horizon. Table 1 defines the inputs. No observations are estimated from LSEG or another data provider. The full comparisons and source/code hashes are retained in `results/capped_gbm`.

The coupled meshes use $M = 128, 256, 512, 1024, 2048, 4096, 8192, 16384$ and $N = \lceil M^2/8 \rceil$. Additional experiments refine time at fixed space and space at fixed time. The per-contract heat boxes contain both kinks and use nine diffusion standard deviations of margin, with a minimum margin of 0.3. These finite experiments keep the box fixed; the whole-line convergence theorem additionally requires its tail allowance to vanish. Domain-enlargement checks preserve both h and k, so a boundary change is not confused with changing the spatial resolution.

The spot bump is $\min\{0.1S, S(0.006h^{1/8} + h)\}$. Volatility, rate and maturity bumps are respectively $\min\{0.25\sigma, 0.10\sigma h^{1/4}\}$, $0.01h^{1/4}$ and $\min\{0.25\tau, 0.05\tau h^{1/4}\}$. All actual steps and signed bump biases are retained per comparison. The analytic central difference uses precisely the PDE bump and therefore isolates its bias. The tridiagonal and sine-basis solvers are compared at one, three and 64 time steps, including large values of λ.

Figure 1 exposes non-monotone refinement rather than imposing a fitted rate. Kink/grid alignment and interpolation can affect price error; differencing amplifies these errors in the Greeks. For the short-dated gamma at the finest mesh, the analytic bump bias is approximately -4.28×10^{-4}, while the PDE difference at that bump is approximately 1.45×10^{-5}. The larger gamma discrepancy is therefore mainly a finite-bump effect. Finer price accuracy alone does not remove it. Independent mesh refinements and all five Greek comparisons remain available even

Quantity	Maximum absolute error at $M = 16384$	Units
Price	9.950×10^{-7}	Monetary units
Delta	1.111×10^{-4}	Per unit spot
Gamma	4.137×10^{-4}	Per unit spot squared
Vega	1.829×10^{-3}	Per unit volatility
Rho	3.171×10^{-4}	Per unit rate
Theta	1.026×10^{-3}	Per year of calendar time

Table 2: Worst absolute discrepancy from the analytic price/Greek across the seven synthetic contracts. Greek errors include the declared bump bias. Different derivative units are not combined into a single accuracy score.

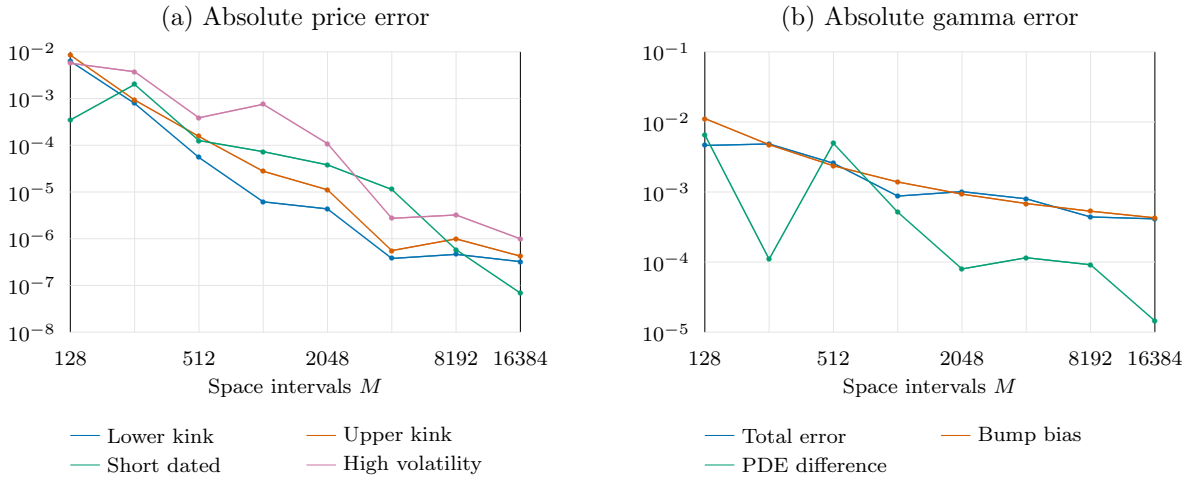


Figure 1: Synthetic convergence evidence. Panel (a) contrasts the lower and upper payoff kinks, a short-dated low-volatility claim and a high-volatility claim. Panel (b) separates the short-dated gamma total error into analytic bump bias and the PDE difference at the same bump. Absolute component errors need not add because their signed contributions can cancel. Both axes are logarithmic. All seven contracts and all five Greeks are retained in the comparison files.

where a selected plot would conceal adverse behaviour.

Every Monte Carlo method receives 262,144 terminal-payoff evaluations per contract. Raw sampling uses that many independent normals. Antithetic sampling uses 131,072 independent pairs. Controlled sampling uses 16,384 independent pilot pairs and 114,688 production pairs, charging both terminal evaluations of each pair. A control coefficient is fitted separately for each output and clipped to $[-10^6, 10^6]$; the pilot is independent of production. Shared production normals between methods are declared, and no confidence claim is made for their paired differences. The seed is 732451, with deterministic case offsets. Terminal-evaluation budgets measure sampling work; arithmetic costs and wall-clock time are not asserted to be identical between estimators.

The 49 raw outputs comprise the price, five Greeks and additional mixed gamma in each of seven contracts. Each receives failure allowance 0.05/49. Their simultaneous finite-sample bounds apply to ideal independent exact normal draws and exact arithmetic. Their floating-point constants and pseudo-random normal generation have not been formally certified. Normal sample-variance intervals are reported separately as pointwise asymptotic diagnostics. Observed inclusion in the conservative bounds is a sanity check, not a proof of coverage.

The bounded mixed estimator reduces measured raw gamma variance in every tested contract. This is an observed comparison, not uniform dominance over model parameters. Con-

Contract	Analytic Γ	Raw LR SE	Raw mixed SE	Variance ratio
Lower kink	4.026×10^{-3}	7.689×10^{-5}	1.966×10^{-5}	15.293
Upper kink	-7.259×10^{-3}	6.210×10^{-5}	2.269×10^{-5}	7.493
Inside spread	-3.183×10^{-3}	7.019×10^{-5}	1.523×10^{-5}	21.240
Deep cap	-2.747×10^{-3}	4.014×10^{-5}	2.021×10^{-5}	3.945
Low-volatility short dated	0.296226	3.320×10^{-3}	8.632×10^{-4}	14.792
Negative rate/dividend	2.713×10^{-3}	4.478×10^{-5}	1.272×10^{-5}	12.400
High volatility	-3.680×10^{-4}	4.251×10^{-6}	2.287×10^{-6}	3.455

Table 3: Raw gamma estimators at equal terminal-evaluation budgets. SE is the measured standard error, not a finite-sample confidence radius. The variance ratio is ordinary payoff-LR estimator variance divided by mixed-estimator variance. All antithetic and independently fitted control variants remain in the full comparison.

trolled ordinary likelihood-ratio gamma remains more efficient in several cases. Stock controls also worsen measured delta and rho efficiency for the interior-spread contract at the matched total budget. The analytic call spread has neither sampling error nor a finite-domain or mesh error and remains the strongest baseline here.

6 Conditional valuation and jointly transformed controls

I use conditional valuation to connect measure-theoretic probability with an implemented variance comparison. Conditional Monte Carlo is established (Asmussen, 2018). The following argument supplies complete proofs and an explicit GBM specialisation; it does not claim a new quantum theory. The expectation and variance in this section are under the stipulated pricing measure Q. For a sub-σ-algebra $\mathcal{G}$, write $P_{\mathcal{G}}Y = \mathbb{E}_Q[Y \mid \mathcal{G}]$. The space $L^2(\mathcal{G})$ consists of equivalence classes admitting a $\mathcal{G}$-measurable square-integrable representative.

Theorem 6.1 (Conditional expectation as an orthogonal projection)**.** *For $Y \in L^2(Q)$, $L^2(\mathcal{G})$ is a closed subspace of $L^2(Q)$ and $m = P_{\mathcal{G}}Y$ is its unique nearest element. Equivalently, $m \in L^2(\mathcal{G})$ and $\mathbb{E}_Q[\mathbf{1}_A m] = \mathbb{E}_Q[\mathbf{1}_A Y]$ for every $A \in \mathcal{G}$. For every $Z \in L^2(\mathcal{G})$,*

$$\mathbb{E}_Q[(Y-Z)^2] = \mathbb{E}_Q[(Y-m)^2] + \mathbb{E}_Q[(m-Z)^2], \tag{30}$$

$$m^2 \le \mathbb{E}_Q[Y^2 \mid \mathcal{G}] \quad Q\text{-almost surely}, \tag{31}$$

$$\mathrm{Var}_Q(Y) = \mathrm{Var}_Q(m) + \mathbb{E}_Q[(Y-m)^2]. \tag{32}$$

For $\mathcal{G}_1 \subseteq \mathcal{G}_2$, the projections satisfy the tower property. The variance of the conditional mean increases with information, while its mean-square prediction error decreases.

The two variance statements concern different objects. Replacing Y by m in an estimator lowers its intrinsic sampling variance. Increasing the retained information makes m closer to Y and consequently raises the variance of m. Appendix D proves the closedness, projection characterisation, conditional inequality and nested-information identities without assuming the result being demonstrated.

The control-variate construction is established (Nelson, 1990). For a centred control vector $C \in L^2(Q)^d$, let $\Sigma_C = \mathbb{E}_Q[CC^{\mathsf{T}}]$, $\kappa_C = \mathbb{E}_Q[C(Y - \mathbb{E}_Q Y)]$ and $\mathcal{J}(\beta) = \mathrm{Var}_Q(Y - \beta^{\mathsf{T}}C)$. Let $C_{\mathcal{G}} = P_{\mathcal{G}}C$, componentwise, and define $\mathcal{J}_{\mathcal{G}}(\beta) = \mathrm{Var}_Q(m - \beta^{\mathsf{T}}C_{\mathcal{G}})$.

Proposition 6.2 (Conditioning and control optimisation)**.** *The covariance matrix Σ_C is positive semidefinite, κ_C lies in its range, and all minimisers of $\mathcal{J}$ are $\Sigma_C^{\dagger}\kappa_C + \ker \Sigma_C$, where $\Sigma_C^{\dagger}$ is the Moore–Penrose inverse. For every fixed coefficient vector,*

$$\mathcal{J}(\beta) = \mathcal{J}_{\mathcal{G}}(\beta) + \mathbb{E}_Q\left[\{Y - m - \beta^{\mathsf{T}}(C - C_{\mathcal{G}})\}^2\right]. \tag{33}$$

Consequently $\min_\beta \mathcal{J}_{\mathcal{G}}(\beta) \le \min_\beta \mathcal{J}(\beta)$. *If* β_{raw} *and* $\beta_{\mathcal{G}}$ *are population minimisers, then*

$$\mathcal{J}_{\mathcal{G}}(\beta_{\mathrm{raw}}) - \mathcal{J}_{\mathcal{G}}(\beta_{\mathcal{G}}) = (\beta_{\mathrm{raw}} - \beta_{\mathcal{G}})^{\mathsf{T}} \Sigma_{C,\mathcal{G}} (\beta_{\mathrm{raw}} - \beta_{\mathcal{G}}), \tag{34}$$
$$\Sigma_{C,\mathcal{G}} = \mathbb{E}_Q[C_{\mathcal{G}} C_{\mathcal{G}}^{\mathsf{T}}].$$

Thus projection is linear at fixed coefficients, but the optimal raw coefficient need not remain optimal after conditioning.

For a terminal claim with horizon T and initial spot $S_0 = S$, set $Y = e^{-rT}F(S_T)$ and choose $t = uT$ with $0 < u < 1$. Independent Brownian increments give

$$m_t = \mathbb{E}_Q[Y \mid \mathcal{F}_t] = \mathbb{E}_Q[Y \mid \sigma(S_t)] = e^{-rt} V(S_t, T-t), \tag{35}$$
$$C = e^{-rT} S_T - S_0 e^{-qT},$$
$$c_t = \mathbb{E}_Q[C \mid \mathcal{F}_t] = e^{-rt-q(T-t)} S_t - S_0 e^{-qT}. \tag{36}$$

The Brownian filtrations $\mathcal{F}_t$ are nested; the individual $\sigma(S_t)$ need not be. This distinction justifies comparing population conditional variances across observation times. In particular,

$$\mathrm{Var}_Q(c_t) = S_0^2 e^{-2qT}(e^{\sigma^2 t} - 1). \tag{37}$$

At $t = 0$, $m_0 = V(S_0, T)$ is the already available analytic price. This zero-variance endpoint requires evaluating that price and is not a new valuation algorithm.

I fit coefficients on a pilot independent of the production sample and bound their norm before reuse. Appendix D.5 proves conditional unbiasedness and finite variance for this procedure. The population inequality in Proposition 6.2 does not guarantee that a separately fitted conditional coefficient wins on a finite production sample. The comparison therefore charges pilot observations, terminal simulation and conditional-price evaluations separately. The call-spread formula remains the strongest price baseline. Unbiasedness here concerns exact projections; evaluating them in floating-point arithmetic is not itself an unbiasedness certificate.

For the quantum basket method, the auxiliary-variance proposition already uses the established total-variance identity. The present projection argument explains that identity as a least-squares result and shows why controls must be transformed jointly with the score. It does not change the earlier finite-law or gate certificates. Reducing intrinsic variance alone proves neither an overall computational advantage nor a quantum advantage: the conditional evaluator and its required accuracy must also be charged.

6.1 Matched-information comparisons and evaluator work

I reuse the seven synthetic contracts in Table 1 and choose $u = 0.25, 0.5, 0.9$. Every method has the same model parameters, known stock-control means and analytic pricing function. The study uses independent pilots of 8,192 observations and production samples of 65,536, with coefficient norm bound 10^6 and seed 840731. Methods share production draws within each split. The split simulations have correct marginal Brownian laws, but different observation times are not coupled into one sampled Brownian path.

I compare raw payoffs, exact conditional valuation, raw pilot controls, conditional valuation with the raw pilot coefficient, separately refitted conditional controls, and the unbiased comparator that conditions the payoff but leaves its control unprojected. The analytic call spread is included in every case as the zero-sampling-error reference. There are 147 comparison rows, including that reference. Thirty-two additional independent pilot fits per case/split give 672 coefficient-risk diagnostics.

Population raw moments use the truncated lognormal formulae derived in Appendix D.7. Conditional moments use one-dimensional Gaussian quadrature at orders 160 and 320. These numerical integrals are diagnostics, not rigorous interval enclosures. The maximum refinement

Contract	Raw optimal	Conditional optimal	Ratio	Unprojected penalty
Lower kink	14.303	1.1166	12.810	24.327
Upper kink	21.919	2.9954	7.318	21.04
Inside spread	17.683	1.5105	11.707	27.345
Deep cap	7.0584	1.1945	5.909	0.71552
Short dated	0.16239	0.036354	4.467	0.25742
Negative rate/dividend	17.781	1.2724	13.975	25.975
High volatility	70.98	5.5678	12.748	44.194

Table 4: Population-variance diagnostics at $t = T/2$, in squared monetary units. Conditional quantities are approximated by order-320 Gaussian quadrature. The ratio is optimal raw-control variance divided by optimal conditional-control variance. The final column is the additional variance from leaving the stock control unprojected at the raw population coefficient. These are oracle comparisons, not finite-pilot efficiency guarantees.

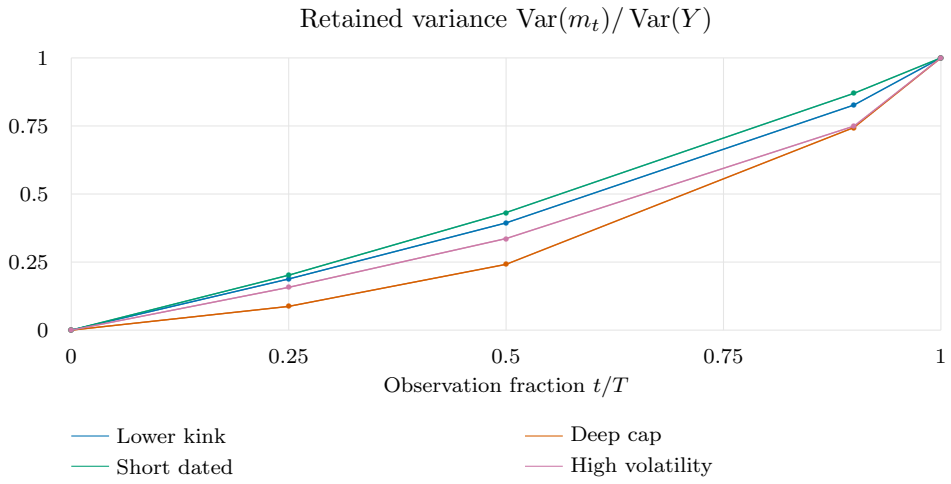


Figure 2: Retained conditional-price variance for selected synthetic contracts. Interior markers use the one-dimensional moment diagnostics; endpoints follow the analytic model. Lines join evaluated points. Increasing the observation time retains more terminal-price randomness: estimator variance rises while payoff-prediction loss falls. These are distinct quantities. All seven contracts remain in the comparison files; the figure reports no computational speed or quantum advantage.

difference across recorded smooth moments is approximately 3.83×10^{-8}; a different representation integrates the analytic residual-life variance to check the projection decomposition. Its maximum discrepancy is approximately 1.21×10^{-10}. Coarse 80×80 quadrature of the kinked joint payoff instead has an identity discrepancy as large as 4.62 and remains an explicitly failed resolution diagnostic. It is not used as validation of the smooth results.

Across the 21 interior cases/splits, the numerical oracle variance ratios range from 1.2866 to 93.7667. They illustrate the population inequality, without proving a universal numerical gain of that magnitude. The conditional pilot fit has greater computed population excess variance than the raw pilot coefficient reused on the projected control in 152 of 672 independent pilot realisations. This is a descriptive diagnostic of coefficient-estimation risk, not a general probability or efficiency claim. The fixed-coefficient projection inequality and the oracle optimum inequality remain separate from this fitted-coefficient comparison.

A stand-alone raw estimate requires one Gaussian variate, one terminal stock exponential and one capped payoff evaluation per observation. A conditional estimate requires one Gaussian variate, one intermediate stock exponential and four normal-CDF evaluations. Its independent

conditional pilot has the same conditional-evaluation requirement. Reusing a raw pilot coefficient charges the raw pilot, while the unprojected-control comparator also needs the additional terminal simulation. Per-policy counts and the actual shared diagnostic generation counts are retained separately; shared code execution is not presented as individual estimator timings.

For a population variance v and per-observation work c, an idealised large-sample work proxy at target mean-square sampling error ϵ^2 is cv/ϵ^2. The numerical variance ratio can therefore identify a break-even production-cost ratio, but it excludes pilots, finite confidence, kernel error and arithmetic. No CDF-to-payoff work conversion or wall-clock superiority is established. The available analytic price still dominates sampling in this model.

The conditional call-spread evaluator uses survival-function differences for intervals in the upper normal tail and avoids subtracting two large stock-valued calls. A range fallback removes tiny floating-point violations: 49 production states, four main-pilot states and 149 repeated-pilot states use it, with largest recorded discounted adjustment 3.46×10^{-15}. Those observations do not bound the complete kernel error or prove numerical unbiasedness. The oracle/projection statements refer to the exact conditional operator; the reported arithmetic and quadrature diagnostics retain their narrower scope.

7 Physical estimation and volatility uncertainty

I now distinguish a stipulated pricing parameter from an estimated one. The estimation experiment uses synthetic ex-dividend observations; it tests statistical reasoning and the propagation of parameter uncertainty, without establishing empirical model adequacy. I retain constant coefficients, known dividends, equally spaced noise-free observations and independent Gaussian increments. These assumptions exclude stochastic volatility, serial dependence and market microstructure noise.

For d assets under P, write $dS_j/S_j = (\mu_j - q_j)\,dt + (L\,dW^P)_j$, where $\Sigma = LL^\top \succ 0$ and μ_j is a total-return drift. For spacing $\delta > 0$, the $n \geq 2$ observed log-return vectors satisfy

$$R_i \sim N_d(a\delta, \Sigma\delta), \qquad a_j = \mu_j - q_j - \tfrac{1}{2}\Sigma_{jj}, \qquad C = \sum_{i=1}^{n}(R_i - \bar{R})(R_i - \bar{R})^\top. \tag{38}$$

Appendix E derives the likelihood and proves

$$\widehat{a} = \bar{R}/\delta, \qquad \widehat{\Sigma}_{\mathrm{ML}} = C/(n\delta), \qquad \widehat{\Sigma}_{\mathrm{U}} = C/((n-1)\delta). \tag{39}$$

The full covariance maximum exists uniquely precisely when $C \succ 0$. When $n \leq d$, its rank is at most $n - 1 < d$ and the likelihood is unbounded over positive-definite covariance matrices. Positive-definite shrinkage therefore changes the estimation rule; it does not recover a nonexistent unconstrained maximum.

The normal-sample laws have established foundations (Student, 1908; Wishart, 1928). Gaussian row rotations give independent sample means and scatters, and the exact Wishart law. They yield marginal Student-t intervals for a_j and chi-squared intervals for $\sigma_j = \sqrt{\Sigma_{jj}}$. The physical total-return estimator $\widehat{\mu}_{\mathrm{U},j} = \widehat{a}_j + q_j + \widehat{\Sigma}_{\mathrm{U},jj}/2$ is unbiased; using the maximum-likelihood covariance instead gives bias $-\Sigma_{jj}/(2n)$. The Student-t interval concerns a_j, not this corrected total-return drift. At fixed observation horizon $T_{\mathrm{obs}} = n\delta$, the variance of $\widehat{a}_j$ is $\Sigma_{jj}/T_{\mathrm{obs}}$: more frequent ideal observations improve covariance precision but do not remove drift uncertainty. The sampling-frequency distinction also has a published finance antecedent (Merton, 1980); the final journal metadata and working-paper reading are distinguished in the source ledger. Under the stipulated constant-diffusion change of measure, covariance is shared by P and Q, while the pricing drift remains $r - q_j$ regardless of the estimated physical drift.

Proposition 7.1 (The image of a volatility confidence interval)**.** *For fixed known S, K, H, τ, r, q, set*

$$c_* = \frac{\log(K(K+H)/S^2)}{\tau} - 2(r-q). \tag{40}$$

If $c_* \leq 0$, the capped price is strictly decreasing in $\sigma > 0$. If $c_* > 0$, it increases up to $\sigma_* = \sqrt{c_*}$ and decreases thereafter. For an interval $[\ell, u] \subset (0, \infty)$, its minimum price is the smaller endpoint price; its maximum is the largest price among the endpoints and σ_* when $\sigma_* \in [\ell, u]$. Consequently, mapping an exact marginal volatility confidence interval through this full image gives a price interval with at least the same coverage under the stipulated model.

The proof uses the sign of vega in (8); no monotonicity assumption is imposed on a capped claim. The synthetic comparison includes an interval straddling σ_*, for which endpoint-only propagation misses the upper price. At a true interior maximum, the exact endpoint-only interval has zero coverage, whereas the full image preserves the volatility interval's coverage. This is a financial consequence of changing convexity across the two strikes. The coverage statement is marginal, conditional on known contractual inputs and the model assumptions; floating-point quantiles and prices are not outward-certified bounds. Shrinkage does not inherit the exact chi-squared pivot, and statistical uncertainty remains distinct from PDE and Monte Carlo numerical errors. These are established estimation principles and a checked application of the call-spread sensitivity, rather than a claim of theoretical priority.

Conditional numerical price guarantees at the interval endpoints and included volatility maximum compose with statistical coverage to give at least $1-\alpha-\beta$ overall coverage (Appendix E). A classical/quantum comparison must share estimation information and charge all required pricing calls; the floating-point experiment does not implement this certificate.

7.1 Synthetic estimation and price-coverage evidence

I generate three-dimensional log returns using physical total drifts $(0.06, 0.04, 0.08)$, dividend yields $(q, 0.015, 0.02)$ and volatilities $(\sigma, 0.8\sigma, 1.2\sigma)$. Here q, σ are the first-asset values of each designed contract. The correlation matrix is

$$\begin{pmatrix} 1 & 0.4 & -0.2 \\ 0.4 & 1 & 0.25 \\ -0.2 & 0.25 & 1 \end{pmatrix}. \tag{41}$$

I retain the seven original contracts and add a first-asset contract with $S = K = 100$, $H = 20$, $\tau = 1$, $r = 0.03$, $q = 0.01$ and $\sigma = \sqrt{\log(1.2) - 0.04} = \sigma_*$. Each case uses 1,000 independent synthetic datasets of 252 observations at spacing $1/252$. Marginal intervals use $\alpha = 0.05$; simultaneous mean/diagonal-variance rectangles allocate $0.05/(2d)$ per interval. The seed is 926341, with fixed case offsets. Code and summaries are retained in `results/capped_gbm/calibration`; raw returns are not saved.

For the volatility-maximum contract, the full image covers 948 of 1,000 datasets; endpoint-only propagation covers none. The pointwise 95 per cent Clopper–Pearson interval for the former coverage frequency is $[0.93236, 0.96092]$, and the zero-success interval has upper endpoint 0.003682 (Clopper and Pearson, 1934). Across all cases the corrected price counts range from 937 to 965; the simultaneous parameter-rectangle counts range from 944 to 962. These fluctuations remain visible. Neither the repeated-data binomial intervals nor the individual parameter intervals are claimed to be simultaneously calibrated across every reported diagnostic. The exact probability proof and numerical coverage check are distinct.

The deterministic counterexample in Figure 3 uses $[0.9\sigma_*, 1.1\sigma_*]$. Its largest endpoint price is 6.81278566, whereas the true interior maximum is 6.82481668. This gap is a missing extremum rather than Monte Carlo sampling error. No statistical confidence level is assigned to that illustrative interval.

Contract	Full price image	Endpoints only	Plug-in price RMSE
Lower kink	937/1000	937/1000	0.0760989
Upper kink	946/1000	946/1000	0.189076
Inside spread	954/1000	954/1000	0.0676302
Deep cap	951/1000	951/1000	0.123294
Short dated	959/1000	959/1000	0.0231021
Negative rate/dividend	965/1000	965/1000	0.08144
High volatility	939/1000	939/1000	0.129354
Volatility maximum	948/1000	0/1000	0.00447807

Table 5: Observed coverage and plug-in error across independent synthetic datasets. Coverage counts concern marginal 95 per cent volatility intervals and their price images. RMSE is in monetary units. Observed counts are numerical diagnostics, not statements that each experiment attained at least 95 per cent coverage.

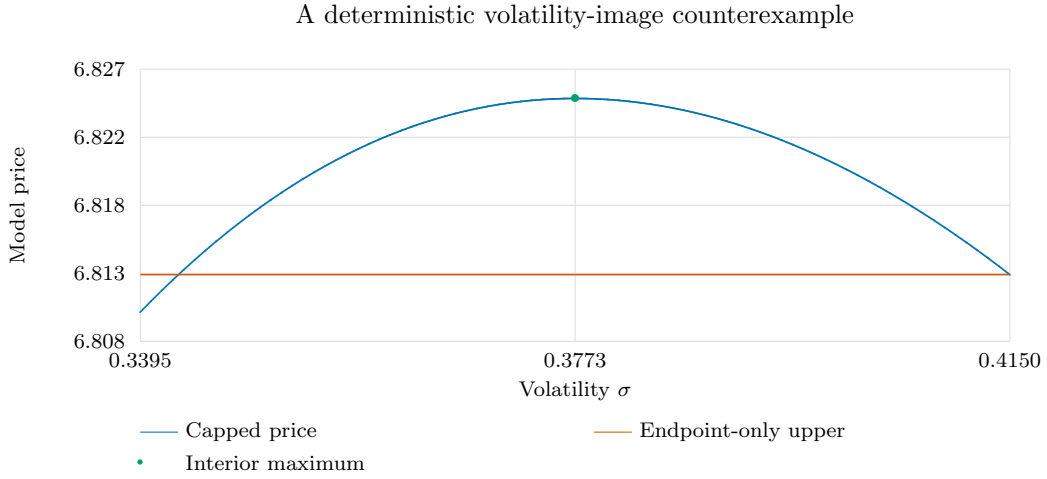


Figure 3: A deterministic interval $[0.9\sigma_*, 1.1\sigma_*]$ for the synthetic volatility-maximum contract. No confidence level is assigned to this illustration. Its maximum price is attained inside the interval, so the endpoint-only upper value misses a legitimate price. The repeated-data confidence experiment uses the chi-squared intervals separately.

The experiment also keeps a singular three-observation/three-asset sample. Its numerical scatter rank is two, and the unrestricted covariance maximum does not exist. Trace-target shrinkage with intensity 0.1 gives positive eigenvalues, but changes the estimator and supplies no transplanted exact interval law. Numerical rank uses floating-point tolerances; the theorem concerns exact rank. The study records estimation bias, covariance error, pivot moments and all interval failures. It estimates a stipulated synthetic model, without validating market prices or identifying empirical volatility risk premia. Pricing uses $r - q$, not a fitted physical return mean.

8 Interpretation and reproducibility

This companion closes a specific preparation gap by connecting an admissible financial strategy, a PDE, a nonsmooth numerical convergence proof and independently checked sensitivities. The bounded mixed-gamma derivation gives a useful estimator and a finite-sample uncertainty statement for this claim. It is an explicit specialisation of sensitivity principles, not a claimed new quantum theory or a completed priority claim. The comparison retains negative gamma/vega, non-monotone errors and adverse control rankings where they occur.

n	Horizon	Spacing	Empirical log-drift SD	Theoretical SD
63	1	1/63	0.20042	0.20000
252	1	1/252	0.19918	0.20000
1008	1	1/1008	0.20298	0.20000
252	1	1/252	0.20573	0.20000
1008	4	1/252	0.10021	0.10000

Table 6: First-asset log-drift uncertainty over 1,000 independent datasets per regime. The repeated one-year daily row belongs to the separate horizon comparison. Increasing frequency at a fixed one-year horizon does not reduce the drift floor; extending the observation horizon does. The result assumes ideal independent diffusion increments.

The numerical pricing comparisons use a stipulated one-asset model with known parameters. The separate estimation experiment fits synthetic observations and propagates parameter uncertainty. It does not validate market prices, solve a general portfolio/control problem or establish a quantum advantage. It also does not extend a one-asset analytic benchmark into a closed-form solution for arbitrary baskets. Those boundaries are essential when using the companion as evidence of financial mathematics and computation.

The engine is `capped_gbm.py`; `run_capped_gbm_study.py` reproduces the comparisons. The tests include active-region Gaussian quadrature of the mixed gamma, analytic sensitivity checks, the direct/sine-basis backward-Euler comparison and projection/singular-control checks. Gaussian inference and price-image checks use `gbm_calibration.py` and `run_calibration_study.py`; their results and source ledger are retained in `results/capped_gbm/calibration`. The conditioning code and source ledger are retained in `results/capped_gbm/conditioning`. The source ledgers record publication details, the versions actually consulted and the precise reading scope. Detailed arguments follow in the appendices. I retain this document as a completed companion study; the main quantum basket paper remains a working paper.

A Financial and analytic derivations

A.1 Change of measure and the self-financing equation

For a concrete construction, suppose under a physical measure P that $dS_t = (\mu - q)S_t\,dt + \sigma S_t\,dW_t^P$ with constant total-return drift μ. Set $\vartheta = (\mu - r)/\sigma$. Novikov's condition follows from $\exp(\vartheta^2 T/2) < \infty$ on every finite horizon. Thus

$$\left.\frac{dQ}{dP}\right|_{\mathcal{F}_T} = \exp\{-\vartheta W_T^P - \vartheta^2 T/2\}, \qquad W_t^Q = W_t^P + \vartheta t \tag{42}$$

defines an equivalent probability measure and yields (2). The stochastic exponential is strictly positive and has expectation one by the Gaussian moment-generating function. For this constant coefficient, the Brownian conclusion can also be checked directly. On any deterministic partition, the density is the product of factors $\exp\{-\vartheta\Delta W^P - \vartheta^2\Delta t/2\}$. Multiplying the independent $N(0,\Delta t)$ increment densities by these factors and completing the square changes each mean to $-\vartheta\Delta t$, leaves its variance Δt and preserves their factorisation. Thus the increments of $W^Q = W^P + \vartheta t$ are independent centred normals with the Brownian variances. The same calculation conditional on the preceding Brownian filtration proves independence from its past. Continuity is inherited from W^P. Hence W^Q is a Brownian motion under Q, an elementary verification of this established constant-coefficient measure change. Consequently this construction does not depend on an unverified calibration of μ.

Let $B_t = e^{rt}$, let δ_t be the share holding and let γ_t be the bank-account holding. A self-

financing wealth $P_t = \delta_t S_t + \gamma_t B_t$ receives asset dividends, so

$$dP_t = \delta_t\, dS_t + q\delta_t S_t\, dt + r(P_t - \delta_t S_t)\, dt = rP_t\, dt + \delta_t \sigma S_t\, dW_t^Q. \tag{43}$$

For an admissible strategy, the share holding is predictable and

$$\int_0^T e^{-2rt}\delta_t^2\sigma^2 S_t^2\, dt < \infty \quad Q\text{-almost surely}, \tag{44}$$

while discounted wealth is bounded below by a deterministic constant. For such a strategy, discounted wealth is a local martingale bounded below and hence a supermartingale. One may see the latter implication by localisation and conditional Fatou's lemma after adding the fixed lower bound. An arbitrage with zero initial wealth and non-negative, sometimes strictly positive terminal wealth would contradict the supermartingale expectation inequality. This proves no arbitrage within this explicit admissible strategy class; it does not assert a general fundamental theorem for arbitrary markets.

A.2 The analytic call and replication

The event $X > A$ is $Z > -d_{2,A}$. Completing the square gives

$$\mathbb{E}[e^{vZ}\mathbf{1}_{\{Z>z\}}] = e^{v^2/2}\Phi(v - z). \tag{45}$$

Substitution in $D\mathbb{E}[(X - A)^+]$ yields (5); payoff linearity yields the call spread. The normal identity

$$Se^{-q\tau}\varphi(d_{1,A}) = AD\varphi(d_{2,A}) \tag{46}$$

follows directly by subtracting the two squared normal arguments. It cancels the derivative-of-Φ terms when differentiating the call formula. The normal arguments have derivatives

$$\begin{aligned}
&\partial_S d_{1,A} = \partial_S d_{2,A} = \frac{1}{Sv}, && \\
&\partial_\sigma d_{1,A} = \sqrt{\tau} - \frac{d_{1,A}}{\sigma}, && \partial_\sigma d_{2,A} = -\frac{d_{1,A}}{\sigma}, \\
&\partial_r d_{1,A} = \partial_r d_{2,A} = \frac{\sqrt{\tau}}{\sigma}, && \\
&\partial_\tau d_{1,A} = \frac{r - q + \sigma^2/2}{\sigma\sqrt{\tau}} - \frac{d_{1,A}}{2\tau}, && \partial_\tau d_{2,A} = \partial_\tau d_{1,A} - \frac{\sigma}{2\sqrt{\tau}}.
\end{aligned} \tag{47}$$

In the spot derivative of C_A, the two density terms cancel exactly, leaving $e^{-q\tau}\Phi(d_{1,A})$; another spot derivative gives $e^{-q\tau}\varphi(d_{1,A})/(Sv)$. In the volatility derivative, their combined contribution is $Se^{-q\tau}\varphi(d_{1,A})(\partial_\sigma d_{1,A} - \partial_\sigma d_{2,A}) = Se^{-q\tau}\varphi(d_{1,A})\sqrt{\tau}$. The rate-argument derivatives cancel, leaving $\tau AD\Phi(d_{2,A})$ from differentiating the discount factor. Finally,

$$\partial_\tau C_A = -qSe^{-q\tau}\Phi(d_{1,A}) + rAD\Phi(d_{2,A}) + \frac{Se^{-q\tau}\sigma\varphi(d_{1,A})}{2\sqrt{\tau}}. \tag{48}$$

Subtracting the corresponding strike-J expressions from those at strike K proves every formula in (6)–(10). The lognormal convolution is smooth at positive maturity, even though F has two kinks. Dominated convergence proves continuity at expiry: $X \to S$ almost surely, F is continuous and $0 \le F \le H$.

To verify the heat transformation explicitly, first write $x = \log(S/K)$ and $V = e^{-r\tau}u(\tau, x)$. The chain rule gives $SV_S = e^{-r\tau}u_x$ and $S^2V_{SS} = e^{-r\tau}(u_{xx} - u_x)$. Equation (4) therefore reduces to $u_\tau = au_{xx} + bu_x$. Setting $u(\tau, x) = w(\tau, x + b\tau)$ gives $w_\tau = aw_{yy}$. Conversely, Gaussian convolution proves (11); reversing the calculation proves (4) without assuming it in the valuation argument.

With calendar time t and expiry T, apply Itô's formula to $V(S_t, T-t)$ on $[0, T-\varepsilon]$. The pricing equation gives

$$dV(S_t, T-t) = rV(S_t, T-t)\,dt + \sigma S_t V_S(S_t, T-t)\,dW_t^Q. \tag{49}$$

This agrees with (43) for $\delta = V_S$ and $\gamma = (V - SV_S)/B$. Furthermore $0 \le V_S \le e^{-q(T-t)}$ by (6), and the Gaussian moment-generating function gives the explicit second moment

$$\mathbb{E}_Q S_t^2 = S_0^2 \exp\{[2(r-q) + \sigma^2]t\}. \tag{50}$$

It is finite and bounded on every finite horizon. Hence $\mathbb{E}_Q \int_0^T e^{-2rt}\sigma^2 S_t^2 V_S^2\,dt < \infty$. The discounted stochastic gain is therefore a true square-integrable martingale. Continuity at expiry gives terminal wealth $F(S_T)$ as $\varepsilon \downarrow 0$. Its discounted wealth is non-negative, so the strategy is admissible. Any bounded quoted claim valuation different from the replication cost permits buying the cheaper position, selling the dearer one and investing the initial difference; their terminal claim cash flows cancel. This gives a strictly positive terminal gain with zero initial wealth and a bounded-below discounted wealth process, contradicting no arbitrage. Any other bounded classical solution with this terminal payoff has a discounted martingale representation with the same conditional terminal expectation; equivalently, apply the stopped Itô argument and bounded convergence. Its price equals (3). This proves the stated uniqueness.

B Numerical stability and convergence proofs

B.1 The matrix maximum principle

Let A_h be the $(M-1)\times(M-1)$ matrix with diagonal $1+2\lambda$ and off-diagonals $-\lambda$. If $A_h z \ge 0$ componentwise and z had a negative minimum at an interior row, that row would satisfy $(A_h z)_j \le z_j < 0$. At an edge row, the missing neighbour is interpreted as zero and gives an even stronger contradiction. Thus $z \ge 0$. The same argument applied to a null vector and its negative proves that A_h is nonsingular; consequently $A_h^{-1} \ge 0$. The row sums of A_h are one in interior rows and $1+\lambda$ at the two edge rows. Therefore $A_h\mathbf{1} \ge \mathbf{1}$, whence

$$0 \le A_h^{-1}\mathbf{1} \le \mathbf{1}, \qquad \|A_h^{-1}\|_\infty \le 1. \tag{51}$$

This proves contraction for two solutions with equal boundary values. For positivity and the upper bound, the actual right-hand side is the previous interior vector plus λH at the last row. If the previous vector lies between 0 and $H\mathbf{1}$, this right-hand side lies between 0 and $A_h(H\mathbf{1})$. Applying the non-negative inverse proves the invariant interval by induction. No condition relating k and h is used.

The discrete eigenvectors are $\sin(j\ell\pi/M)$ and substitution in the centred second difference gives $-\nu_\ell$. The affine lift $\ell(y) = H(y-L)/(U-L)$ has zero discrete second difference. Each lifted mode is consequently multiplied by $(1+ak\nu_\ell)^{-1}$ per step, which proves (14) after N steps. This identity is in exact arithmetic; the direct-step comparison checks implementation error.

B.2 Mollification without a false smooth-payoff assumption

Let $\rho_\eta(z) = \eta^{-1}\rho(z/\eta)$ and $f_\eta = f * \rho_\eta$. Global Lipschitz continuity gives

$$\|f_\eta - f\|_\infty \le L_f m_1 \eta, \qquad \|f_\eta^{(4)}\|_\infty \le L_f \|\rho_\eta'''\|_1 = L_f c_3 \eta^{-3}. \tag{52}$$

The second estimate uses the weak first derivative f', which exists almost everywhere and is bounded by L_f. Moving the three remaining derivatives to the smooth kernel proves the formula and its bound. Since the radius is smaller than the distance of either kink to the boundary, f_η equals 0 near L and H near U.

Subtract the affine lift from f_η and shift the domain to $[0, U-L]$. The lifted function is zero at both endpoints and affine in their neighbourhoods. Its even derivatives, of order at least two, vanish there. Odd reflection at both endpoints therefore produces a smooth function of period $2(U-L)$: the value and every derivative agree at each reflection. In particular, the fourth derivative of the extension has maximum norm at most $L_f c_3 \eta^{-3}$. Convolve this extension with the whole-line Gaussian heat kernel and add the affine lift back. Oddness keeps the lifted solution zero at both boundaries, so this is precisely the finite-domain smooth solution u_η. Gaussian convolution contracts the maximum norm and commutes with derivatives; hence

$$\sup_{0\le s\le\tau} \|\partial_y^4 u_\eta(s,\cdot)\|_\infty \le L_f c_3 \eta^{-3}, \qquad \partial_s^2 u_\eta = a^2 \partial_y^4 u_\eta. \tag{53}$$

At time $s_n = nk$, insert $u_\eta(s_n, y_j)$ into the numerical equation and divide its residual by k. The integral remainder for backward time differencing is at most $k \sup|\partial_s^2 u_\eta|/2$. The integral or Taylor remainder for the centred second difference is at most $h^2 \sup|\partial_y^4 u_\eta|/12$. Thus the residual R_j^n obeys

$$|R_j^n| \le \left(\frac{a^2 k}{2} + \frac{a h^2}{12}\right) L_f c_3 \eta^{-3}. \tag{54}$$

If the smooth grid starts from f_η, its error satisfies $A_h e^n = e^{n-1} - kR^n$ with $e^0 = 0$. Equation (51) and induction give $\|e^N\|_\infty \le Nk \sup_n \|R^n\|_\infty$. Both the continuous maximum principle and the discrete contraction bound the change from f_η back to f by $L_f m_1 \eta$. Adding these two changes proves (16). Taking η proportional to $(h^2+k)^{1/4}$, eventually below the fixed boundary margin, proves convergence and the stated upper rate. This is a proof for the actual kinked initial condition, not for a replacement smooth contract.

B.3 Domain truncation and interpolation

Let $Y_s = x + \sigma B_s$, and let ζ be its first exit from $[L, U]$. The stopped Brownian representation of the finite-domain solution is

$$w^{\mathrm{box}}(\tau, x) = \mathbb{E}_x[f(Y_\tau)\mathbf{1}_{\{\zeta>\tau\}}] + \mathbb{E}_x[g(Y_\zeta)\mathbf{1}_{\{\zeta\le\tau\}}], \qquad g(L) = 0, \quad g(U) = H. \tag{55}$$

It follows either from the stopped Itô formula for positive-time smooth solutions and bounded convergence, or from the killed heat semigroup. The full solution is $\mathbb{E}_x f(Y_\tau)$. On exit, the terminal payoff and boundary value are both in $[0, H]$, so their difference is bounded in absolute value by H. The reflection principle bounds each one-sided exit event by twice its corresponding normal tail; a union bound proves (17). The upper and lower exit contributions can have opposite signs, so this is an absolute error bound rather than an ordering of the two prices.

At positive maturity, differentiate the whole-line convolution once using the bounded weak derivative f', then once using the Gaussian density. For a Gaussian with standard deviation v, the L^1 norm of its density derivative is $\sqrt{2/\pi}/v$. Therefore

$$\|w_{yy}(\tau,\cdot)\|_\infty \le \frac{L_f}{\sigma\sqrt{\tau}}\sqrt{\frac{2}{\pi}}. \tag{56}$$

Linear interpolation on an interval of width h has error at most $h^2 \sup|w_{yy}|/8$. Interpolation weights are non-negative and sum to one. Apply the domain bound at the two bracketing nodes and the nodal discretisation bound to these same weights; add the whole-line interpolation error and discount. This proves (18), without requiring a new boundary regularity bound for w^{box}.

B.4 Uniform consistency for Greeks

For a central first difference, price error contributes at most $(E+E)/(2d) = E/d$. Taylor's theorem with an integral remainder gives bias at most $M_3 d^2/6$. For the central second difference,

price error contributes at most $(E + 2E + E)/d^2 = 4E/d^2$, and the fourth-order remainder is at most $M_4 d^2/12$. These prove (19) and (20).

On a compact parameter set with positive lower bounds for S, σ and τ, the analytic formula is smooth and its third and fourth parameter derivatives are bounded. Its transformed queries also lie in a compact interval. Let $L \to -\infty$ and $U \to \infty$, keep $h \to 0$ and $k \to 0$, and use a radius proportional to $(h^2 + k)^{1/4}$. The Gaussian exit tails vanish uniformly because $\sigma\sqrt{\tau}$ has a finite upper bound, and the other terms in (18) vanish uniformly. The discount factors are bounded even when r is negative. Thus $E_n \to 0$ uniformly. Choosing the bump sequence as stated in Proposition 4.1 proves Greek convergence. A fixed domain or a fixed price error does not meet this argument merely because its bumps are made smaller.

C Monte Carlo derivations and uncertainty proofs

C.1 Pathwise first derivatives

The capped payoff is Lipschitz in its asset argument, with derivative $\mathbf{1}_{\{K<x<J\}}$ except at two points. The continuously distributed X hits those points with probability zero. Its parameter derivatives are

$$X_S = X/S, \qquad X_\sigma = X(\sqrt{\tau}Z - \sigma\tau), \qquad X_r = \tau X, \qquad X_\tau = X\left(b + \frac{\sigma Z}{2\sqrt{\tau}}\right). \tag{57}$$

On any compact parameter neighbourhood with $S, \sigma, \tau > 0$, difference quotients of the Lipschitz payoff are dominated by a constant times $(1 + |Z|)\exp(C|Z|)$, which is integrable against a Gaussian. This follows from the mean-value theorem for the smooth terminal map X and the Lipschitz bound for F. Dominated convergence therefore justifies interchanging differentiation and expectation. Differentiating the discount factor as well gives (21) and (22). Neither payoff kink contributes a first-order term because the payoff itself is continuous there.

On the active region, $X \le J$ and $Z \in (z_-, z_+)$, giving the delta, vega and theta ranges in (26). For rho the expression $IX - F(X)$ is exactly 0, K or $-H$ in the below-strike, active and capped regions, respectively. This proves its stated tighter range.

C.2 The likelihood-ratio gamma

For $x > 0$, the lognormal density is

$$p_S(x) = \frac{1}{xv\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{\log(x/S) - b\tau}{v}\right)^2\right]. \tag{58}$$

Writing $z = (\log(x/S) - b\tau)/v$, direct differentiation gives

$$\frac{\partial_S p_S(x)}{p_S(x)} = \frac{z}{Sv}, \qquad \frac{\partial_S^2 p_S(x)}{p_S(x)} = \frac{1}{S^2}\left(\frac{z^2 - 1}{v^2} - \frac{z}{v}\right). \tag{59}$$

On a compact spot neighbourhood away from zero, the density and its first two spot derivatives admit a common envelope

$$C(1 + |\log x|^2)x^{-1}\exp\{-|\log x|^2/(4v^2)\}, \qquad x > 0, \tag{60}$$

for a finite constant C. Indeed $\log S + b\tau$ is bounded on the neighbourhood and $(u - m)^2 \ge u^2/2 - m^2$; the derivative weights are polynomials of degree at most two in $u = \log x$, with bounded inverse spot factors. The envelope is integrable by the substitution $u = \log x$, and

remains integrable after multiplication by $F \le H$. Two applications of dominated convergence to $D \int F(x) p_S(x)\, dx$ prove unbiasedness of (23). Using $F \le H$ gives

$$\begin{aligned}\mathbb{E}[Y_\Gamma^2] &\le \frac{D^2 H^2}{S^4} \mathbb{E}\left[\left(\frac{Z^2-1}{v^2} - \frac{Z}{v}\right)^2\right] \\ &= \frac{D^2 H^2}{S^4}\left(\frac{2}{v^4} + \frac{1}{v^2}\right), \end{aligned} \tag{61}$$

because $\mathbb{E}(Z^2-1)^2 = 2$, $\mathbb{E}Z^2 = 1$ and the odd mixed moment is zero. Variance is no larger than the second moment, proving (28). Gamma's likelihood-ratio estimator is unbounded; a bounded-payoff concentration inequality cannot be applied to it without this separate argument.

C.3 A bounded mixed gamma rival

The pathwise delta has the fixed-asset representation

$$\Delta = \frac{D}{S}\int_K^J x p_S(x)\, dx. \tag{62}$$

The integration limits are fixed when spot varies. On this bounded interval away from zero, $x\partial_S p_S(x)$ has a common integrable envelope on every compact spot neighbourhood. Differentiation is therefore legitimate even though the payoff derivative is an indicator. Using (59),

$$\begin{aligned}\partial_S \Delta &= -\frac{D}{S^2}\int_K^J x p_S(x)\, dx + \frac{D}{S}\int_K^J x \partial_S p_S(x)\, dx \\ &= \frac{D}{S^2}\int_K^J x p_S(x)\left(\frac{z}{v} - 1\right) dx. \end{aligned} \tag{63}$$

This proves unbiasedness of (24). It includes the boundary effect through the density derivative. On the active region, $X < J$ and $|Z| \le m$, proving (29) and its finite second moment. As an independent analytic check, Gaussian tilting reduces its expectation to $e^{-q\tau}\{\varphi(d_{1,K}) - \varphi(d_{1,J})\}/(Sv)$, in agreement with (7). I combine the established pathwise and likelihood-ratio constructions of Broadie and Glasserman (1996), deriving the capped-payoff expression explicitly here; I make no priority claim. The ordinary and mixed gamma estimators are compared using identical financial parameters and matched terminal evaluation budgets; neither estimator is discarded after seeing its variance.

C.4 Confidence guarantees and stronger classical sampling

For completeness, the bounded-variable concentration argument of Hoeffding (1963) can be checked directly. If Y lies in an interval of width $R > 0$, define $A(s) = \log \mathbb{E}\exp\{s(Y - \mathbb{E}Y)\}$ for real s. Boundedness justifies differentiating twice. The second derivative is the variance of Y under the exponentially tilted probability law. That law has the same interval support; centring at its midpoint gives $A''(s) \le R^2/4$. Since $A(0) = A'(0) = 0$, integration twice yields

$$A(s) \le R^2 s^2/8, \qquad s \in \mathbb{R}. \tag{64}$$

For independent observations and $s > 0$, exponential Markov's inequality and the product moment-generating function therefore give

$$\mathbb{P}(\overline{Y} - \mathbb{E}Y \ge t) \le \exp\{n(R^2 s^2/8 - st)\}. \tag{65}$$

The minimiser is $s = 4t/R^2$, giving the upper-tail bound $\exp(-2nt^2/R^2)$. Apply the same argument to $-Y$ and sum the two tail bounds. Solving $2\exp(-2nt^2/R^2) = \alpha$ proves (27). If $R =$

0, the observation is constant and its confidence radius is zero. Independence gives $\operatorname{Var}(\overline{Y}_\Gamma) \leq B_\Gamma^2/n$; Chebyshev's inequality gives the stated gamma radius. A union bound requires no independence between different outputs calculated on the same paths. Their allocated failure probabilities must sum to the intended familywise level.

For the stock control $C = DX - Se^{-q\tau}$, the Gaussian exponential moment gives $\mathbb{E}C = 0$ and a finite second moment. For any estimator Y, a deterministic coefficient β yields

$$\operatorname{Var}(Y - \beta C) = \operatorname{Var}(Y) - 2\beta \operatorname{Cov}(Y, C) + \beta^2 \operatorname{Var}(C). \tag{66}$$

When $\operatorname{Var}(C) > 0$, the population minimiser is $\beta_* = \operatorname{Cov}(Y, C)/\operatorname{Var}(C)$ and the minimum variance is $\operatorname{Var}(Y) - \operatorname{Cov}(Y, C)^2/\operatorname{Var}(C)$. An estimated pilot coefficient need not attain that minimum. Independence of pilot and main sample preserves conditional unbiasedness; a zero pilot denominator is handled by selecting zero control coefficient. Antithetic pairs replace $Y(Z)$ and $C(Z)$ with their averages at Z and $-Z$ and preserve their expectations, but variance reduction for every Greek is not guaranteed merely by symmetry. The comparison therefore reports measured variance and complete sampling work. For fixed finite pilot coefficients, the main-pair estimators have finite variance, so their studentised normal intervals have asymptotic validity under non-zero variance. Finite-sample exact coverage is not claimed.

D Conditional projection and control-variate proofs

D.1 Completeness, closedness and the nearest-point construction

For this appendix write $\mathbb{E} = \mathbb{E}_Q$, $\langle X, Z\rangle = \mathbb{E}[XZ]$ and $\|X\|_2 = (\mathbb{E}[X^2])^{1/2}$. Random variables are identified when they agree Q-almost surely. All variables are real-valued. First, $L^2(Q)$ is complete. From any Cauchy sequence (X_n) choose a subsequence (X_{n_j}) with $\|X_{n_{j+1}} - X_{n_j}\|_2 \leq 2^{-j}$. Minkowski's inequality and Fatou's lemma give

$$\left\| \sum_{j \geq k} |X_{n_{j+1}} - X_{n_j}| \right\|_2 \leq \sum_{j \geq k} 2^{-j}. \tag{67}$$

The non-negative infinite sum is consequently finite almost surely. The subsequence converges pointwise to a measurable finite limit X, with the value zero assigned on its null non-convergence set. The same tail bound proves $X \in L^2(Q)$ and $X_{n_j} \to X$ in L^2. The Cauchy property then gives $X_n \to X$ in L^2. If each X_n admits a $\mathcal{G}$-measurable representative, the convergence set and pointwise limit of this subsequence are $\mathcal{G}$-measurable. Hence X admits a $\mathcal{G}$-measurable representative. This proves that $L^2(\mathcal{G})$ is closed; linearity of that space follows directly from measurability and the triangle inequality.

Fix $Y \in L^2(Q)$ and set $d = \inf_{Z \in L^2(\mathcal{G})} \|Y - Z\|_2$. Choose (Z_n) whose distances converge to d. The midpoint lies in the subspace, so the parallelogram identity yields

$$\|Z_n - Z_k\|_2^2 = 2\|Y - Z_n\|_2^2 + 2\|Y - Z_k\|_2^2 - 4\left\|Y - \frac{Z_n + Z_k}{2}\right\|_2^2 \leq 2\|Y - Z_n\|_2^2 + 2\|Y - Z_k\|_2^2 - 4d^2. \tag{68}$$

Thus (Z_n) is Cauchy. Completeness and closedness give a limit $m \in L^2(\mathcal{G})$ attaining d. For each $h \in L^2(\mathcal{G})$, the quadratic

$$\|Y - m - sh\|_2^2 = \|Y - m\|_2^2 - 2s\langle Y - m, h\rangle + s^2\|h\|_2^2 \tag{69}$$

has its minimum at $s = 0$, implying $\langle Y - m, h\rangle = 0$. Conversely this orthogonality proves (30), and therefore minimality. If two minimisers exist, their difference is in the subspace and orthogonal to itself, so they agree almost surely.

D.2 The integral characterisation and conditional Jensen inequality

Taking $h = \mathbf{1}_A$ in the orthogonality relation proves the displayed indicator-integral characterisation in Theorem 6.1. Conversely, that characterisation first implies orthogonality against every $\mathcal{G}$-measurable simple function by finite linearity. For any $h \in L^2(\mathcal{G})$, truncate h to $[-n, n]$ and then approximate that truncation by a simple function with pointwise error at most $1/n$. The resulting simple sequence converges to h in L^2, since $\mathbb{E}[h^2 \mathbf{1}_{\{|h|>n\}}] \to 0$ by dominated convergence. Cauchy–Schwarz then extends orthogonality to h. Thus the integral characterisation is equivalent to the projection construction. It is also the usual definition of conditional expectation. Its uniqueness among integrable $\mathcal{G}$-measurable candidates follows by testing the difference on its positive and negative sets.

The conditional expectation of Y^2 requires an L^1 extension rather than an assumption that $Y \in L^4$. The projection preserves positivity: if $U \geq 0$, $U \in L^2$, the integral identity on $A = \{P_{\mathcal{G}} U < 0\}$ contradicts a negative integral unless $Q(A) = 0$. Linearity and this positivity show that the projection preserves order. For $U \geq 0$, $U \in L^1$, define $h_n = P_{\mathcal{G}}(U \wedge n)$, choosing simultaneous representatives with $0 \leq h_n \leq h_{n+1}$. Monotone convergence gives

$$\begin{aligned} h = \lim_n h_n < \infty \text{ almost surely}, \qquad \mathbb{E}h = \mathbb{E}U, \\ \mathbb{E}[\mathbf{1}_A h] = \mathbb{E}[\mathbf{1}_A U] \quad (A \in \mathcal{G}). \end{aligned} \tag{70}$$

The almost-sure finiteness follows because $\mathbb{E}h < \infty$. Assigning zero on the $\mathcal{G}$-measurable null set where the limit is infinite gives an integrable representative. Positive and negative parts define conditional expectation for every L^1 variable. The integral characterisation proves linearity and positivity of this extension by uniqueness.

For every $A \in \mathcal{G}$, $\mathbf{1}_A m \in L^2(\mathcal{G})$, so orthogonality implies $\mathbb{E}[\mathbf{1}_A Y m] = \mathbb{E}[\mathbf{1}_A m^2]$. Expanding the square on A therefore proves

$$\mathbb{E}[(Y - m)^2 \mid \mathcal{G}] = \mathbb{E}[Y^2 \mid \mathcal{G}] - m^2 \geq 0 \quad \text{almost surely.} \tag{71}$$

Here products are integrable by Cauchy–Schwarz, and the equality follows from the L^1 integral characterisation on every A. This proves conditional Jensen in the required square case. Taking expectations proves $\|P_{\mathcal{G}} Y\|_2 \leq \|Y\|_2$. Linearity follows from the indicator-integral identity and uniqueness; idempotence follows because the projection fixes every element of its subspace. Taking $Z = \mathbb{E}Y$ in (30), and using $\mathbb{E}m = \mathbb{E}Y$, proves the total-variance identity.

D.3 Nested information and the direction of variance comparisons

Let $m_i = P_{\mathcal{G}_i} Y$ for $\mathcal{G}_1 \subseteq \mathcal{G}_2$. Every indicator of $\mathcal{G}_1$ is also an indicator of $\mathcal{G}_2$; applying the integral characterisation twice gives $P_{\mathcal{G}_1} m_2 = m_1$. Also $P_{\mathcal{G}_2} m_1 = m_1$ by measurability. The first identity is the tower property. Orthogonality against $m_2 - m_1$ gives

$$\operatorname{Var}(m_2) = \operatorname{Var}(m_1) + \mathbb{E}[(m_2 - m_1)^2], \tag{72}$$

$$\mathbb{E}[(Y - m_1)^2] = \mathbb{E}[(Y - m_2)^2] + \mathbb{E}[(m_2 - m_1)^2]. \tag{73}$$

For the first equality, the cross term vanishes by projecting $m_2 - m_1$ onto $\mathcal{G}_1$. For the second, $Y - m_2$ is orthogonal to the $\mathcal{G}_2$-measurable difference $m_2 - m_1$. These exact identities distinguish dispersion of the conditional mean from the prediction loss of approximating the terminal variable by it.

D.4 Multivariate controls, singular covariance and optimal coefficients

Let C have d centred square-integrable components and let $\mu = \mathbb{E}Y$. The matrix $\Sigma_C = \mathbb{E}[CC^{\mathsf{T}}]$ is symmetric and positive semidefinite, since $v^{\mathsf{T}} \Sigma_C v = \mathbb{E}[(v^{\mathsf{T}} C)^2] \geq 0$. If $v \in \ker \Sigma_C$, this square

has zero expectation and $v^\mathsf{T}C = 0$ almost surely. Therefore $v^\mathsf{T}\kappa_C = \mathbb{E}[(v^\mathsf{T}C)(Y-\mu)] = 0$. For a finite symmetric matrix, its spectral decomposition gives $\operatorname{range}\Sigma_C = (\ker\Sigma_C)^\perp$, so $\kappa_C \in \operatorname{range}\Sigma_C$. Writing $\Sigma_C = U\operatorname{diag}(\lambda_i)U^\mathsf{T}$, define $\Sigma_C^\dagger = U\operatorname{diag}(d_i)U^\mathsf{T}$, where $d_i = 1/\lambda_i$ when $\lambda_i > 0$ and $d_i = 0$ when $\lambda_i = 0$. Then $\beta_* = \Sigma_C^\dagger\kappa_C$ satisfies $\Sigma_C\beta_* = \kappa_C$. Direct square expansion gives

$$\begin{aligned}\mathcal{J}(\beta) &= \operatorname{Var}(Y) - 2\beta^\mathsf{T}\kappa_C + \beta^\mathsf{T}\Sigma_C\beta \\ &= \operatorname{Var}(Y) - \kappa_C^\mathsf{T}\Sigma_C^\dagger\kappa_C + (\beta-\beta_*)^\mathsf{T}\Sigma_C(\beta-\beta_*). \end{aligned} \tag{74}$$

All minimisers are exactly $\beta_* + \ker\Sigma_C$. A singular coefficient vector is not uniquely identifiable, but its controlled random variable is unique almost surely because null directions produce zero controls.

Componentwise conditional projection preserves centring and square-integrability of C. By linearity, $P_\mathcal{G}(Y - \beta^\mathsf{T}C) = m - \beta^\mathsf{T}C_\mathcal{G}$ for every fixed β. Its residual $Y - m - \beta^\mathsf{T}(C - C_\mathcal{G})$ is orthogonal to $L^2(\mathcal{G})$. Applying total variance proves (33). Minimising first on the conditional side, then evaluating at a raw population minimiser, proves

$$\min_\beta \mathcal{J}_\mathcal{G}(\beta) \le \mathcal{J}_\mathcal{G}(\beta_{\mathrm{raw}}) \le \mathcal{J}(\beta_{\mathrm{raw}}) = \min_\beta \mathcal{J}(\beta). \tag{75}$$

Applying (74) to the projected variables proves (34), including singular $\Sigma_{C,\mathcal{G}}$.

Optimisation does not generally commute with conditioning. For an explicit example, let A, B be independent symmetric random signs, $Y = A + B$, $C = A + 2B$ and $\mathcal{G} = \sigma(A)$. Then $m = A$, $C_\mathcal{G} = A$, $\beta_{\mathrm{raw}} = 3/5$ and $\beta_\mathcal{G} = 1$. The respective variances are $\mathcal{J}(3/5) = 1/5$, $\mathcal{J}_\mathcal{G}(3/5) = 4/25$ and $\mathcal{J}_\mathcal{G}(1) = 0$. The square penalty is exactly $4/25$. For nested information and projected controls, applying (72) to the fixed terminal residual $Y - \beta^\mathsf{T}C$ shows $\mathcal{J}_{\mathcal{G}_1}(\beta) \le \mathcal{J}_{\mathcal{G}_2}(\beta)$ for every β. Taking infima gives the same non-decreasing order for the population-optimised residual variance.

Conditioning the response while leaving the control untransformed is a different estimator. Since m and $C_\mathcal{G}$ are $\mathcal{G}$-measurable, orthogonality gives

$$\operatorname{Var}(m - \beta^\mathsf{T}C) = \mathcal{J}_\mathcal{G}(\beta) + \beta^\mathsf{T}(\Sigma_C - \Sigma_{C,\mathcal{G}})\beta. \tag{76}$$

The extra term is non-negative. Both estimators are unbiased for a fixed coefficient, because both controls have mean zero; the distinction here is a variance penalty rather than a bias assertion.

D.5 Independent pilots and the limits of the oracle comparison

Let $\mathcal{P}$ be the pilot σ-algebra, independent of the production pairs $(Y_i, C_i)_{i=1}^n$, which are independent copies of (Y, C). Choose a $\mathcal{P}$-measurable coefficient $\widehat{\beta}$ with $\|\widehat{\beta}\| \le B < \infty$ almost surely. For example, project a fitted coefficient onto a deterministic Euclidean ball. This restriction supplies moment control even when an empirical covariance matrix is nearly singular. Define

$$\widehat{\mu} = n^{-1}\sum_{i=1}^n (Y_i - \widehat{\beta}^\mathsf{T}C_i). \tag{77}$$

Each summand is square-integrable, because $(Y_i - \widehat{\beta}^\mathsf{T}C_i)^2 \le 2Y_i^2 + 2B^2\|C_i\|^2$. Independence fixes the production distribution when conditioning on $\mathcal{P}$, while centring gives $\mathbb{E}[C_i \mid \mathcal{P}] = 0$. Conditional independence therefore proves

$$\mathbb{E}[\widehat{\mu} \mid \mathcal{P}] = \mu, \tag{78}$$

$$\operatorname{Var}(\widehat{\mu} \mid \mathcal{P}) = n^{-1}\mathcal{J}(\widehat{\beta}), \qquad \operatorname{Var}(\widehat{\mu}) = n^{-1}\mathbb{E}[\mathcal{J}(\widehat{\beta})]. \tag{79}$$

The unconditional variance follows from total variance; the variance of the conditional mean is zero. The same argument applies to $(m_i, C_{\mathcal{G},i})$ if the production projections are exact. For any such pilot,

$$\mathrm{Var}(\widehat{\mu}) = \frac{\mathrm{Var}(Y) - \kappa_C^{\mathsf{T}} \Sigma_C^{\dagger} \kappa_C}{n} + \frac{\mathbb{E}[(\widehat{\beta} - \beta_*)^{\mathsf{T}} \Sigma_C (\widehat{\beta} - \beta_*)]}{n}. \tag{80}$$

This non-negative excess-risk term explains why separately estimated raw and conditional coefficients need not preserve the oracle ranking. If the same pilot coefficient is applied to both jointly projected production variables, however, (33) holds conditionally on the pilot and its population variance order survives averaging over that pilot. None of these arguments applies automatically to coefficients fitted on the production sample: mean-zero controls alone do not make a data-dependent product have expectation zero. Pilot observations remain part of the computational budget. Normal intervals based on production sample variance are asymptotic, not finite-sample guarantees. Specifically, with a fixed finite pilot held constant while $n \to \infty$ and $\mathcal{J}(\widehat{\beta}) > 0$ for almost every pilot, the conditional central limit theorem and consistency of the production sample variance give a studentised normal limit. The finite second moments proved above suffice for these two statements; no fourth moment is required. A zero-variance endpoint is excluded from this studentised limit.

If numerical projections $\widetilde{m}, \widetilde{C}_{\mathcal{G}}$ obey the almost-sure deterministic bounds $|\widetilde{m} - m| \le \varepsilon_m$ and $\|\widetilde{C}_{\mathcal{G}} - C_{\mathcal{G}}\| \le \varepsilon_C$, then their pilot-controlled mean satisfies

$$\left| \mathbb{E}\left[n^{-1} \sum_i (\widetilde{m}_i - \widehat{\beta}^{\mathsf{T}} \widetilde{C}_{\mathcal{G},i}) \right] - \mu \right| \le \varepsilon_m + B\varepsilon_C. \tag{81}$$

Indeed subtract the exact estimator, apply the triangle and Cauchy–Schwarz inequalities to each summand, then take expectations. Without a validated evaluator-error bound, the floating-point computation does not inherit an exact numerical unbiasedness guarantee from the projection theorem.

D.6 GBM conditional valuation, stock controls and endpoint limits

Under (2), for $0 \le t < T$,

$$S_T = S_t \exp\{b(T-t) + \sigma(W_T^Q - W_t^Q)\}. \tag{82}$$

The increment is independent of $\mathcal{F}_t$ and normally distributed with variance $T-t$. Integrating the bounded payoff against its normal density gives (35). The resulting expression is measurable with respect to $\sigma(S_t)$. Testing indicators of that smaller σ-algebra shows that it is also $\mathbb{E}[Y \mid \sigma(S_t)]$. This equality for the terminal payoff does not imply that the spot-generated σ-algebras are nested.

The Gaussian exponential moment gives $\mathbb{E}[S_T \mid \mathcal{F}_t] = S_t e^{(r-q)(T-t)}$, proving (36) and $\mathbb{E}c_t = 0$. Moreover $c_t = S_0 e^{-qT}\{\exp(\sigma W_t^Q - \sigma^2 t/2) - 1\}$, so

$$\begin{aligned} \mathrm{Var}(C) &= S_0^2 e^{-2qT}(e^{\sigma^2 T} - 1), \\ \mathbb{E}[(C - c_t)^2] &= S_0^2 e^{-2qT}(e^{\sigma^2 T} - e^{\sigma^2 t}). \end{aligned} \tag{83}$$

The first two moments of the exponential and total variance prove these expressions and (37). Since $0 \le Y, m_t \le e^{-rT} H$, continuity of the GBM paths and continuity of V at positive remaining maturity imply $m_t \to V(S_0, T)$ in L^2 as $t \downarrow 0$ by dominated convergence. Continuity at expiry gives $m_t \to Y$ in L^2 as $t \uparrow T$. Equation (83) gives the corresponding L^2 limits for the controls. The nested-filtration identities therefore apply through both endpoint limits. They concern prices under the fixed model, not parameter derivatives of a simulation coupling; no conditional Greek identity is asserted here.

D.7 Analytic raw moments and smooth conditional diagnostics

Let $D_T = e^{-rT}$, $J = K + H$, $\theta = \mathbb{E}Y$ and $s_C = S_0 e^{-qT}$. For $\ell = 0, 1, 2$ and $0 \le A < B \le \infty$, define

$$\begin{aligned} M_\ell(A,B) &= \mathbb{E}[S_T^\ell \mathbf{1}_{\{A<S_T<B\}}] \\ &= S_0^\ell e^{\ell bT + \ell^2\sigma^2 T/2}\left[\Phi\left(\frac{\log(B/S_0) - bT - \ell\sigma^2 T}{\sigma\sqrt{T}}\right) - \Phi\left(\frac{\log(A/S_0) - bT - \ell\sigma^2 T}{\sigma\sqrt{T}}\right)\right]. \end{aligned} \tag{84}$$

Zero and infinite endpoints use the corresponding normal-tail limits. To derive the expression, substitute $S_T = S_0 e^{bT+\sigma\sqrt{T}Z}$, complete the square in $e^{\ell\sigma\sqrt{T}z}\varphi(z)$ and shift the normal integration endpoints by $\ell\sigma\sqrt{T}$. Thus

$$\mathbb{E}[Y^2] = D_T^2\{M_2(K,J) - 2KM_1(K,J) + K^2 M_0(K,J) + H^2 M_0(J,\infty)\}, \tag{85}$$

$$\mathrm{Cov}(Y,C) = D_T^2\{M_2(K,J) - KM_1(K,J) + HM_1(J,\infty)\} - \theta s_C. \tag{86}$$

These follow by splitting the terminal stock into below-strike, active and capped regions and expanding the respective payoff polynomials. Subtracting θ^2 gives $\mathrm{Var}(Y)$, while the stock-control variance is already derived in (83).

For conditional diagnostics, let $s(z) = S_0 e^{bt+\sigma\sqrt{t}z}$, $m(z) = e^{-rt}V(s(z), T-t)$ and $c(z) = e^{-rt-q(T-t)}s(z) - s_C$. Their moment integrals are

$$v_t = \int_{\mathbb{R}} (m(z)-\theta)^2\varphi(z)\,dz, \qquad \kappa_t = \int_{\mathbb{R}} (m(z)-\theta)c(z)\varphi(z)\,dz. \tag{87}$$

For an independently expressed projection check, condition the residual $Y - \beta C$ on $S_t = s$. Put $R = T - t$ and $X_R = s\exp(bR + \sigma\sqrt{R}Z')$, where Z' is independent standard normal. The exact inner variance is

$$\mathrm{Var}(Y - \beta C \mid S_t = s) = e^{-2rt}\,\mathrm{Var}\{e^{-rR}F(X_R) - \beta e^{-rR}X_R\}. \tag{88}$$

The constant βs_C does not affect variance. Applying the raw truncated-moment formulae with spot s and horizon R, then integrating the displayed quantity against the density of S_t, evaluates the residual term in (33). This changes the representation being checked and avoids coarse quadrature of the payoff kinks. Its floating-point numerical agreement does not certify an integration remainder.

E Gaussian estimation and confidence-image proofs

E.1 Likelihood, covariance geometry and existence

I assume the observations in (38) are independent and the true Σ is strictly positive definite. For $a \in \mathbb{R}^d$ and any candidate $A \succ 0$, the joint density is

$$(2\pi\delta)^{-nd/2}(\det A)^{-n/2}\exp\left\{-\frac{1}{2\delta}\sum_{i=1}^{n}(R_i - a\delta)^\top A^{-1}(R_i - a\delta)\right\}. \tag{89}$$

The identity

$$\sum_i (R_i - a\delta)(R_i - a\delta)^\top = C + n(\bar{R} - a\delta)(\bar{R} - a\delta)^\top \tag{90}$$

follows by expansion, since $\sum_i (R_i - \bar{R}) = 0$. For every $A \succ 0$, the second term makes the negative log-likelihood strictly larger unless $a = \bar{R}/\delta$. Thus this is the unique maximising mean. After substituting it, minimising the negative log-likelihood is equivalent to minimising

$g(A) = \log\det A + \operatorname{tr}(A^{-1}B)$, where $B = C/(n\delta)$. If $B \succ 0$, let $D = A^{-1/2}BA^{-1/2}$ with eigenvalues $\lambda_1, \dots, \lambda_d > 0$. Trace cyclicity and determinant multiplicativity give

$$g(A) - g(B) = \operatorname{tr} D - \log\det D - d = \sum_{j=1}^{d} (\lambda_j - \log\lambda_j - 1) \geq 0. \tag{91}$$

The scalar derivative is $1 - 1/\lambda$, so each summand has its unique minimum zero at one. Equality therefore requires $D = I$, equivalently $A = B$. This proves a unique global maximum. The parameter transformation $\mu = a+q+\operatorname{diag}(A)/2$ is bijective, so the joint physical-drift/covariance maximiser is unique as well.

If C has rank $k < d$, orthogonal diagonalisation gives $C = U\operatorname{diag}(c_1, \dots, c_k, 0, \dots, 0)U^\top$, with $c_j > 0$. Choose $A_\varepsilon = U\operatorname{diag}(c_1/(n\delta), \dots, c_k/(n\delta), \varepsilon, \dots, \varepsilon)U^\top$. Then $\operatorname{tr}(A_\varepsilon^{-1}B) = k$, while $\log\det A_\varepsilon$ contains $(d-k)\log\varepsilon$. As $\varepsilon \downarrow 0$, the log-likelihood tends to $+\infty$. The case $k = 0$ uses $A_\varepsilon = \varepsilon I$ and the same argument. There is no finite maximiser in the open positive-definite parameter space. The centred $n \times d$ observation matrix has row sum zero, so $\operatorname{rank} C \leq \min(d, n-1)$. In particular, $n \leq d$ forces this failure. Under the Gaussian model, $C \succ 0$ almost surely when $n > d$: after the rotation below, its factor is a $(n-1) \times d$ matrix of independent normal entries. A $d \times d$ minor is a nonzero polynomial in continuously distributed entries; its zero set has Lebesgue measure zero by induction on the number of variables, using finite univariate root sets and Fubini for nonzero coefficient polynomials.

E.2 Orthogonal rotation, exact pivots and drift uncertainty

Let Z be an $n \times d$ matrix of independent standard normals, and write the observation matrix as $R = \delta\mathbf{1}a^\top + \sqrt{\delta}ZL^\top$. Choose an orthogonal $n \times n$ matrix O whose first row is $\mathbf{1}^\top/\sqrt{n}$. The joint density of Z is proportional to $\exp(-\|Z\|_F^2/2)$. The transformation $Z \mapsto OZ$ preserves the Frobenius norm and has absolute Jacobian one. Hence every entry of OZ is again an independent standard normal. Denote its rows by $z_0^\top, z_1^\top, \dots, z_{n-1}^\top$. Taking the first rotated row and removing that row from the centred matrix yields

$$\widehat{a} = a + \frac{Lz_0}{\sqrt{n\delta}}, \qquad \frac{C}{\delta} = L\left(\sum_{i=1}^{\nu} z_i z_i^\top\right)L^\top, \qquad \nu = n-1. \tag{92}$$

These two quantities are independent. I define $W_d(\Sigma, \nu)$ as the distribution of $\sum_{i=1}^{\nu} X_i X_i^\top$ for independent $X_i \sim N_d(0, \Sigma)$. Equation (92) therefore proves $C/\delta \sim W_d(\Sigma, \nu)$, including the singular case $\nu < d$. Since $\mathbb{E}[X_i X_i^\top] = \Sigma$,

$$\mathbb{E}\widehat{\Sigma}_{\mathrm{U}} = \Sigma, \qquad \mathbb{E}\widehat{\Sigma}_{\mathrm{ML}} = \frac{n-1}{n}\Sigma, \qquad \operatorname{Cov}(\widehat{a}) = \frac{\Sigma}{n\delta}. \tag{93}$$

For one coordinate, put $s_j^2 = \widehat{\Sigma}_{\mathrm{U},jj}$. The ν projections $(Lz_i)_j/\sqrt{\Sigma_{jj}}$ are independent standard normals, so

$$Q_j = \frac{\nu s_j^2}{\Sigma_{jj}} \sim \chi_\nu^2, \qquad T_j = \frac{\sqrt{n\delta}(\widehat{a}_j - a_j)}{s_j} \sim t_\nu. \tag{94}$$

Here χ_ν^2 is a sum of ν independent squared standard normals, and t_ν is the ratio of an independent standard normal to $\sqrt{Q_j/\nu}$. Both laws are continuous, so their quantiles give exact marginal coverage. For $0 < \alpha < 1$, let $x_{\nu,p}$ and $t_{\nu,p}$ be their respective p quantiles. Solving the central pivot inequalities for the unknown parameters gives

$$a_j \in \left[\widehat{a}_j - t_{\nu,1-\alpha/2}\frac{s_j}{\sqrt{n\delta}},\ \widehat{a}_j + t_{\nu,1-\alpha/2}\frac{s_j}{\sqrt{n\delta}}\right], \tag{95}$$

$$\sigma_j \in \left[\sqrt{\frac{\nu s_j^2}{x_{\nu,1-\alpha/2}}},\ \sqrt{\frac{\nu s_j^2}{x_{\nu,\alpha/2}}}\right]. \tag{96}$$

The square-root transformation is increasing on positive values; it changes neither event nor coverage. These are marginal intervals. Assigning error probability $\alpha/(2d)$ to each of the d mean and d volatility intervals and applying the union bound gives simultaneous coverage at least $1-\alpha$, without assuming independence across assets. This covers no off-diagonal covariance interval. Joint bounds $a_j \in [a_j^-, a_j^+]$ and $\Sigma_{jj} \in [v_j^-, v_j^+]$ also imply $\mu_j \in [a_j^- + q_j + v_j^-/2, a_j^+ + q_j + v_j^+/2]$; thus the same simultaneous event supplies physical-drift bounds.

Gaussian integration by parts gives $\mathbb{E}Z^{2m} = (2m-1)\mathbb{E}Z^{2m-2}$, hence $\mathbb{E}Z^2 = 1$ and $\mathbb{E}Z^4 = 3$. More generally, differentiating $\mathbb{E}e^{t^\top X} = e^{t^\top \Sigma t/2}$ four times at zero gives $\mathbb{E}[X_i X_j X_k X_l] = \Sigma_{ij}\Sigma_{kl} + \Sigma_{ik}\Sigma_{jl} + \Sigma_{il}\Sigma_{jk}$. Differentiation is justified by the integrability of a polynomial in $\|X\|$ times $e^{c\|X\|}$ for each finite c, obtained by completing the Gaussian square. The independent Wishart summands then give

$$\operatorname{Cov}(\widehat{\Sigma}_{\mathrm{U},ij}, \widehat{\Sigma}_{\mathrm{U},kl}) = \frac{\Sigma_{ik}\Sigma_{jl} + \Sigma_{il}\Sigma_{jk}}{\nu}, \qquad \operatorname{Var}(s_j^2) = \frac{2\Sigma_{jj}^2}{\nu}. \tag{97}$$

The physical-drift estimators are

$$\widehat{\mu}_{\mathrm{U},j} = \widehat{a}_j + q_j + \tfrac{1}{2}s_j^2, \tag{98}$$

$$\widehat{\mu}_{\mathrm{ML},j} = \widehat{a}_j + q_j + \tfrac{1}{2}\widehat{\Sigma}_{\mathrm{ML},jj}. \tag{99}$$

Their moments follow from (93):

$$\mathbb{E}\widehat{\mu}_{\mathrm{U},j} = \mu_j, \tag{100}$$

$$\mathbb{E}\widehat{\mu}_{\mathrm{ML},j} = \mu_j - \frac{\Sigma_{jj}}{2n}, \tag{101}$$

$$\operatorname{Var}(\widehat{\mu}_{\mathrm{U},j}) = \frac{\Sigma_{jj}}{n\delta} + \frac{\Sigma_{jj}^2}{2\nu}. \tag{102}$$

The variance uses the mean/scatter independence, not an assumed zero correlation. The random variance correction makes a Student-t interval for a_j inapplicable directly to μ_j. For fixed $T_{\mathrm{obs}} = n\delta$, the first variance term is $\Sigma_{jj}/T_{\mathrm{obs}}$, whereas the covariance estimator's relative standard deviation is $\sqrt{2/\nu}$. Thus increasingly frequent observations within this ideal diffusion model leave a nonzero drift-estimation floor. With fixed δ and increasing n, these variances tend to zero. Equation (97) also makes every covariance-entry variance vanish, proving mean-square consistency of the covariance matrix, mean and corrected drift in fixed dimension.

For any $A \succeq 0$, $0 < \lambda \leq 1$ and $\eta > 0$, put $A_\lambda = (1-\lambda)A + \lambda\eta I$. Quadratic forms give

$$\lambda_{\min}(A_\lambda) \geq \lambda\eta, \qquad \operatorname{cond}_2(A_\lambda) \leq 1 + \frac{(1-\lambda)\|A\|_2}{\lambda\eta}. \tag{103}$$

Indeed the smallest eigenvalue is at least $\lambda\eta$ and the largest is at most $(1-\lambda)\|A\|_2 + \lambda\eta$; their ratio proves the bound. The implemented trace-target rule takes

$$\widehat{\eta} = \operatorname{tr}(\widehat{\Sigma}_{\mathrm{ML}})/d, \qquad \widehat{\Sigma}_{\mathrm{tr}} = (1-\lambda)\widehat{\Sigma}_{\mathrm{ML}} + \lambda\widehat{\eta} I. \tag{104}$$

On positive sample trace, the same deterministic bound holds with $A = \widehat{\Sigma}_{\mathrm{ML}}$ and $\eta = \widehat{\eta}$. Since $\|A\|_2 \leq \operatorname{tr} A$ for $A \succeq 0$, the trace-target condition number is also at most $1 + d(1-\lambda)/\lambda$. The trace is positive almost surely under the model for $n \geq 2$; all observations being identical has probability zero. A zero trace therefore remains an explicitly rejected deterministic input. Linearity and (93) give

$$\mathbb{E}\widehat{\Sigma}_{\mathrm{tr}} = \frac{n-1}{n}\left[(1-\lambda)\Sigma + \lambda\frac{\operatorname{tr}\Sigma}{d} I\right]. \tag{105}$$

Subtracting Σ gives its bias. Consistency of the raw estimator and continuity of the trace imply convergence to $(1-\lambda)\Sigma + \lambda \operatorname{tr}(\Sigma) I/d$. For fixed $\lambda > 0$, this differs from Σ unless the latter is

spherical. Shrinkage is an established alternative (Ledoit and Wolf, 2004). This rule is neither the unrestricted MLE nor the Ledoit–Wolf selected shrinkage rule, and its diagonals inherit no exact Wishart pivot. It is a positivity remedy, not an inference improvement established by the experiment.

The pricing law uses a different drift. Since L is invertible, set $\theta = L^{-1}(\mu - r\mathbf{1})$. On a finite horizon T, the density $\exp(-\theta^\top W_T^P - \|\theta\|^2 T/2)$ has expectation one and is strictly positive. Completing the squares in the independent Gaussian increment densities, as in Appendix A, makes $W^Q = W^P + \theta t$ Brownian under the equivalent Q. Substitution gives $dS_j/S_j = (r - q_j)dt + (L\,dW^Q)_j$. The diffusion covariance remains $LL^\top = \Sigma$ in this stipulated model; this is not evidence that general physical and pricing volatility parameters coincide empirically.

For completeness, define $\Gamma(x) = \int_0^\infty u^{x-1}e^{-u}\,du$ for $x > 0$. Integration by parts gives $\Gamma(x+1) = x\Gamma(x)$. Normalisation of the ν-dimensional Gaussian in polar coordinates gives sphere area $2\pi^{\nu/2}/\Gamma(\nu/2)$; substituting $q = \|z\|^2$ then gives

$$f_{\chi^2_\nu}(q) = \frac{q^{\nu/2-1}e^{-q/2}}{2^{\nu/2}\Gamma(\nu/2)}, \qquad q > 0. \tag{106}$$

For the independent normal ratio, substituting $z = t\sqrt{q/\nu}$ with Jacobian $\sqrt{q/\nu}$ into the joint density and integrating q using the Gamma integral gives

$$\begin{aligned} f_{t_\nu}(t) &= \frac{1}{\sqrt{2\pi\nu}\,2^{\nu/2}\Gamma(\nu/2)} \int_0^\infty q^{(\nu+1)/2-1}e^{-q(1+t^2/\nu)/2}\,dq \\ &= \frac{\Gamma((\nu+1)/2)}{\sqrt{\nu\pi}\Gamma(\nu/2)}(1+t^2/\nu)^{-(\nu+1)/2}, \qquad t \in \mathbb{R}. \end{aligned} \tag{107}$$

E.3 The capped price image and coverage

Let $J = K + H$ and $h = \log(J/K) > 0$. Then $d_{1,K} - d_{1,J} = h/(\sigma\sqrt{\tau}) > 0$, while

$$d_{1,K} + d_{1,J} = \frac{\log(S^2/(KJ)) + 2(r-q)\tau + \sigma^2\tau}{\sigma\sqrt{\tau}} = \frac{\tau(\sigma^2 - c_*)}{\sigma\sqrt{\tau}}. \tag{108}$$

Since $\varphi(x) = (2\pi)^{-1/2}e^{-x^2/2}$, $\varphi(d_{1,K}) - \varphi(d_{1,J})$ has the opposite sign to $d_{1,K}^2 - d_{1,J}^2$. Factoring the latter as the difference times the sum and using the strictly positive difference shows

$$\operatorname{sgn} V_\sigma = \operatorname{sgn}(c_* - \sigma^2). \tag{109}$$

The prefactor in (8) is strictly positive. Thus if $c_* \le 0$ the price is strictly decreasing. If $c_* > 0$, it is strictly increasing below $\sqrt{c_*}$ and strictly decreasing above it, with exactly one stationary point, a strict global maximum. There is no interior minimum in either case.

The analytic price is continuous on every compact positive volatility interval $I = [\ell, u]$. Its range is an interval by the intermediate value theorem, and its extrema are attained by compactness. The sign calculation proves

$$V(I) = \left[\min\{V(\ell), V(u)\},\ \max\{V(s) : s \in \{\ell, u\} \cup \{\sqrt{c_*} : c_* > 0,\ \sqrt{c_*} \in I\}\}\right]. \tag{110}$$

If the stationary point lies strictly between the endpoints, strict monotonicity gives $V(\sqrt{c_*}) > \max\{V(\ell), V(u)\}$. An endpoint-only upper bound therefore omits a legitimate price. For any random volatility interval $I(R)$ with $P\{\sigma \in I(R)\} = 1 - \alpha$, the deterministic set inclusion $\{\sigma \in I(R)\} \subseteq \{V(\sigma) \in V(I(R))\}$ proves at least $1 - \alpha$ price coverage. Equality need not hold because the price map may be non-injective. If the true $\sigma = \sqrt{c_*}$, its strict global maximum belongs to $V(I)$ precisely when $\sigma \in I$, so the corrected coverage is exactly $1 - \alpha$. In contrast, the endpoint-only interval covers this true price only if an endpoint equals σ. For the chi-squared interval in (96), each endpoint is a strictly increasing continuous function of the positive,

continuously distributed s_j^2, so that event has probability zero. This exact counterexample does not rely on a small numerical discrepancy. The claim assumes all other price inputs known, a correctly specified sampling law and the stated equality of diffusion covariance across measures.

Exact pivot coverage does not certify machine calculations of quantiles or extrema. If an interval is outward-enclosed and each required price evaluation has proved absolute error at most ε, subtracting ε from the smallest computed endpoint value and adding ε to the largest computed candidate value encloses the image, provided the stationary candidate is included or its uncertainty is enclosed as well. A floating-point experiment without those bounds supplies a numerical check, not an exact finite-sample computational certificate.

E.4 Composition with conditional numerical price guarantees

In this subsection, P denotes the joint law of the physical data and the pricing procedure's conditional randomisation, with the stipulated physical data law as its marginal. The target $V(s)$ remains the Q-expectation in (3).

Let D denote the data, let $I(D) = [\ell, u] \subset (0, \infty)$ cover the true volatility with probability at least $1 - \alpha$, and let $E(D) = \{\ell, u\}$. Let $C(D)$ add $\sqrt{c_*}$ when positive and contained in $I(D)$. The candidates depend only on data and known contractual inputs. For each candidate, assume a numerical pricing procedure satisfies, for almost every data realisation,

$$P\{|\widehat{V}(s) - V(s)| > \varepsilon_s \mid D\} \le \beta_s, \qquad \sum_{s \in C(D)} \beta_s \le \beta, \quad \varepsilon_s, \beta_s \ge 0. \tag{111}$$

The allowances may depend on D. A conditional union bound makes simultaneous numerical success have probability at least $1 - \beta$ given D, hence also unconditionally by total expectation. Define

$$\widehat{L} = \min_{s \in E(D)} (\widehat{V}(s) - \varepsilon_s), \qquad \widehat{U} = \max_{s \in C(D)} (\widehat{V}(s) + \varepsilon_s). \tag{112}$$

The lower bound uses only endpoints because they contain the true image minimum; the upper bound uses every extremum candidate. On simultaneous numerical success, Equation (110) gives $V(I(D)) \subseteq [\widehat{L}, \widehat{U}]$. Another union bound yields

$$P\{V(\sigma) \in [\widehat{L}, \widehat{U}]\} \ge 1 - \alpha - \beta. \tag{113}$$

Neither step assumes independence. The certificate requires prices at at most three distinct volatility values per fixed contract. Classical/quantum comparisons must share interval/candidate information, declare confidence allowances and charge all pricing calls and estimation work. The premises require valid conditional guarantees, including numerical input/evaluation errors; the floating-point study does not establish them. This standard composition gives no Greek/basket extension or new quantum algorithm.

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